

# The Cosmochrony Research Programme: A Structural Roadmap and Paper Inventory

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## Abstract

The Cosmochrony programme develops a pre-geometric framework in which physical structure — spacetime, quantum mechanics, gauge symmetry, and Standard Model observables — arises from a single primitive: the local structure of admissible non-injective transitions between observable states. The corpus comprises 71 papers across three theory branches. Branch I (Foundation track) builds the axiomatic base: four axioms suffice to derive the Heisenberg group  $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$  and its Weil representation as theorems, without any background substrate or relaxation mechanism. Branch II (O-series) develops the spectral admissibility sub-programme, deriving and numerically validating the transfer chain  $c_\chi \rightarrow \delta_{\text{pair}} \rightarrow \beta^* \approx 0.126$  connecting the BI saturation constant to the charged-lepton mass hierarchy. Branch III (Q-series and companion papers) derives quantum mechanics, spacetime geometry, gauge structure, and further physical observables from Branch I axioms and Branch II spectral data. Paper Q3 completes the  $\text{SU}(2)$  quantum sector: by combining BI indiscernibility with Schur’s lemma on the Clebsch–Gordan decomposition of  $\chi_{2j+1} \otimes \chi_{2j+1}$ , the universal singlet correlator  $E(\hat{a}, \hat{b}) = -j(j+1)/3 \cdot (\hat{a} \cdot \hat{b})$  and the Born rule are derived for all five admissible sectors of  $2I$ , closing the general- $j$  extension that was open in Q2. The geometric emergence sub-programme (Q5a–Q10, U1, W1, H2) closes the effective spacetime identification: under the Q5a continuum-limit hypotheses, the lifting hypothesis [H-lift] is proved (Q9), asymptotic  $\mathfrak{su}(2)$ -isotropy  $A_H \rightarrow 2$  is proved (Q10, U1), the admissibility weight convergence [H-w] is proved (W1), yielding the effective Lorentzian co-metric  $g^{\mu\nu} = \text{diag}(-2, 2, 2, 2) \propto \eta^{\mu\nu}$  ( $A_\tau = 2$ , Q11 [86]). Q5a v2.0 establishes that [H1] is not needed in full generality. Q5a-O2 [84] proves spectral atomicity of the admissible sector (each pair spans three pure Fourier modes) and closes hypotheses [H- $\mathcal{E}1$ ] and [C] without Nash inequalities. Hypothesis [H2] (strong convergence of the rescaled Weil generators) is proved (H2 paper), closing all hypotheses of the Q5a large- $q$  convergence theorem unconditionally on the admissible sector. O31 develops the structural framework for the  $\text{SU}(3)$  identification: the metaplectic intertwining of  $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$  forces triplet co-admissibility unconditionally for the colour-adapted Cayley graph, and the single-frequency BI fingerprint mechanism (v1.5) makes [H-color] exact pointwise on the standard graph at finite  $q$ , so  $\text{SU}(3)$  and the Standard Model composite  $G_{\text{SM}} = \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$  are identified unconditionally for the standard graph. This paper provides a complete inventory, a logical dependency map, and a structured account of what is proved, structural, heuristic, or open. It is intended both as an entry point for external readers and as an internal navigation reference.

**Keywords.** Non-injective projection; admissibility; Born–Infeld constraint; spectral admissibility; Heisenberg group; Weil representation; emergent spacetime; lepton mass hierarchy; quantum mechanics; gauge structure; programme overview.

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# 1 Introduction and Purpose

The Cosmochrony programme was initiated with a white paper [15] positing that all physical structure emerges from a non-injective projection  $\Pi : \Omega \rightarrow \mathcal{O}$  from a pre-geometric substrate  $\chi$  onto observable space. The initial framework postulated a Born–Infeld (BI) saturation constraint, a relaxation mechanism, and an iterated projection cascade  $U_{t+1} = \Pi \circ \sigma(U_t)$  as the engine of dynamics.

Over the course of 2026, the programme has evolved in two parallel directions. On one side, the spectral admissibility sub-programme (O-series, papers O1–O28) has progressively built the computational machinery for extracting physical observables from the spectral structure of the substrate, culminating in the proof of the unconditional transfer chain in O24 [72]. On the other side, the Foundation paper [6] has refounded the entire framework from four axioms (A1–A4), dispensing with any explicit substrate  $\chi$  and relaxation mechanism: these emerge as consequences of the axiomatic structure rather than being postulated. The Q-series then builds quantum mechanics,  $SU(2)$  symmetry, and Lorentzian spacetime on top of the Foundation axioms and O-series spectral data.

The corpus comprises 71 papers with complex mutual dependencies. A reader entering the programme faces two difficulties: it is not obvious where to start, and the distinction between a theorem, a structural argument, and an open conjecture is not always transparent from individual paper titles. This roadmap addresses both difficulties.

**Conventions on result status.** Throughout this paper each result is classified as one of the following: **proved** (formally proved as a theorem within the stated assumptions); **structural** (derived from structural arguments that are internally consistent but not yet fully formalised); **heuristic** (physically motivated and numerically supported but not yet rigorously derived); **open** (explicitly conjectured or known to be unsettled). These classifications follow the original papers; this roadmap introduces no new claims.

## 2 Conceptual Architecture

The programme is organised in three structural levels.

**Level 1 — Axiomatic primitive.** The primitive is the local structure of admissible non-injective transitions. No background spacetime, no quantum postulate, and no substrate field are introduced at this level. The four axioms are: local projective admissibility (A1), structural non-injectivity (A2), admissibility as non-premature selection (A3), and projection locking (A4). From A1–A2 alone, irreversibility and the arrow of time follow. From A3, the Heisenberg non-commutativity and the Weil representation emerge. From A4, the discreteness of quantum transitions follows. Papers at this level: ENI [34], Foundation [6], HeisenbergStructure, noscale [32].

**Level 2 — Spectral fibre structure.** The admissible fibre is proved to carry the Weil representation  $V_\rho$  of  $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$  for a prime  $q$  (Foundation Theorem 5.6; detailed proof in HeisenbergStructure [33]). The O-series computes the spectral properties of this fibre — in particular the capacity exponent  $\delta_{\text{pair}}$  and its transfer to the cascade exponent  $\beta^*$  governing the charged-lepton mass hierarchy. Papers at this level: SpectralAdmissibility [52], SpectralCapacity [53], SpectralGramRigidity [37], SpectralRelaxation [8], and O1–O31 ([43]–[85]).

**Level 3 — Physical observables.** Effective spacetime geometry, gauge structure, Standard Model charges and masses, gravitational waves, and cosmological observables are derived as consequences of Levels 1–2. No additional physical postulates are introduced beyond the continuum-limit hypotheses of Q5a–Q5b. Papers at this level: the Q-series (Q1 [22], Q2 [46], Q3 [91],

Q5a–Q7), the gravity track (Spectral2 [48], Gravity [25], Thermodynamics, Lorentz [13]), and the companion papers ([10, 12, 2, 14, 24, 3, 70]).

### 3 Branch I: Axiomatic Foundations

#### 3.1 The White Paper

The white paper [15] (“Cosmochrony: A Non-Injective Projection Theory of Emergent Physics,” v3.2) is the programme’s primary overview document. It introduces the substrate field  $\chi$ , the non-injective projection  $\Pi$ , the BI saturation constraint, the relaxation mechanism, and the qualitative derivation of spacetime and quantum structure. The spectral entropy functional  $S_\Pi[g] = \frac{1}{2} \log \det' A_g$  appears here as the structural origin of the Einstein equation.

Version 3.2 incorporates several structural refinements informed by the O-series and Foundation results:

- *Oriented state machine*: the admissible cascade is reformulated as a directed graph on admissible projections; physical histories are directed paths, and the first transition is guaranteed by non-commutativity  $[X, \sigma(X)] = Z \neq 0$  [6].
- *Initial conditions dissolved*: the minimal state  $\chi_0$  cannot be a global minimum of the cascade; the fibre non-commutativity prevents any admissible projection from having trivial kernel, so the cascade begins as a structural necessity.
- *Inaccessibility of absolute zero*: the Heisenberg structure of the admissible fibre forbids projective stasis  $U_{n+1} = U_n$  at any stage.
- *Algebraic necessity of fluctuations (Remark)*: no admissible projection can have trivial kernel, so quantum vacuum fluctuations are a structural necessity, not a contingent feature.
- *Schrödinger as Mosco limit*: the continuous Schrödinger equation is derived as the Mosco limit of the discrete oriented admissibility cascade.
- *Extended numerical scope*: the PTO monotonicity is verified for  $q \in \{29, 61, 101, 151, 211\}$ , with 274 conjugate pairs and 360 post-burn-in steps.

*Role*: Programme overview; structural motivation; first formulation of the BI constraint and the relaxation cascade. *Evolution note*: The explicit substrate  $\chi$  and relaxation mechanism of the white paper are not postulated in the Foundation reformulation: they emerge from axioms A1–A4. The white paper remains the primary narrative reference; Branch I papers are the formal foundation.

#### 3.2 Non-Injectivity as a Structural Necessity of Genuine Emergence

The ENI paper [34] proves the following no-go theorem. Any genuinely emergent (non-isomorphic effective) physical description is necessarily non-injective and intrinsically compressive. The argument is purely structural: given a surjective  $\Pi : \Omega \rightarrow \mathcal{O}$  such that  $\mathcal{O}$  carries no independent microstate-resolving structure beyond  $\Pi$ , non-isomorphism of levels forces non-injectivity. A projection entropy  $S_\Pi \geq 0$  is defined, and the equivalence genuine emergence  $\Leftrightarrow \Pi$  non-injective  $\Leftrightarrow S_\Pi > 0$  is established (**proved**, Theorem 2.1 and Corollary 3.1).

*Role*: Foundational justification for structural non-injectivity (Axiom A2). Version 2 adds Corollary 6 (recursive non-injectivity: any tower of genuinely distinct emergent levels carries one independent non-injective projection per stratum) and Section 5 (towers of emergence, structural colour confinement as projective necessity). A new Remark (algebraic route) shows that in frameworks whose fibre carries a Heisenberg algebra  $[X, Y] = Z \neq 0$ ,  $S_\Pi > 0$  also follows directly

from fibre non-commutativity: a trivial-kernel projection would require simultaneous diagonalisation of  $X$  and  $Y$ , which is structurally forbidden. The two derivations are complementary: Theorem 2.1 answers *what* is forced (framework-independently); the algebraic route answers *why* in Heisenberg-fibred frameworks. Cited by Foundation, HeisenbergStructure [33], Q1 [22], Bell, O32 [77].

### 3.3 Foundation: Admissible Non-Injective Transitions as the Primitive

The Foundation paper [6] (v1.13) establishes the minimal axiomatic basis of the programme. From the four axioms A1–A4 alone it derives: (i) irreversibility and the arrow of time (A1+A2); (ii) the proto-state as a physically real unresolved configuration retaining phase coherence (A3); (iii) a non-trivial commutator between conjugate generators, forced by A3 and the minimality of A1; (iv) the identification of the admissible fibre symmetry group as  $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$  for a prime  $q$ , together with BI parity (A1–A3); (v) the identification  $F_n \cong V_\rho$  (Weil representation) as a theorem (Theorem 5.6); (vi) the discrete character of quantum transitions from A4; (vii) projective incompleteness (Corollary to Theorem 5.4): every admissible projection  $\Pi_n$  has a non-trivial kernel, so no observable description is ever complete — a direct consequence of  $[X, \sigma(X)] = Z \neq 0$ . The paper also establishes the threefold role of  $\Pi_n$  (Remark 3.5): admissibility filter, generator of temporal order, and revelation operator — three roles that are algebraically inseparable and all sourced in the non-commutativity. v1.10 adds the co-metric result  $g^{\mu\nu} = \text{diag}(-2, 2, 2, 2)$  (from Q5b, Q6b, Q8, Q10, Q11) to the results table, and a new section placing the four axioms relative to established formalisms (Hamilton–Jacobi, symplectic geometry, WKB, functional renormalisation group). v1.13 revises the entanglement framework: fibres are now defined as *infra-projectable cells* (unsplit configurations below the resolution of  $\Pi$ ), the derivation of entanglement entropy is integrated into the foundation text by importing the admissible pair entropy of Q3 [91] and ENT-E2 [71], and the Bell non-applicability argument is strengthened: in the projective framework Bell factorisability does not apply at the proto-state level (the joint object is below  $\Pi$ -resolution), recovering the Bell-paper’s conclusion as a structural consequence of A2.

*Status:* **proved** for the algebraic identification of the fibre; **structural** for the continuum limit (deferred to Q5a–Q5b). *Role:* Axiomatic spine. Cited by all Q-series papers, HeisenbergStructure [33], noscale, and recent companion papers.

### 3.4 Non-Factorisability and the Emergence of Heisenberg Structure

HeisenbergStructure [33] provides the detailed proof of Foundation Theorem 5.6, omitted from the Foundation paper for concision. The central structural lemma — *admissible non-factorisability* — is a direct formalisation of Axiom A3 and requires no input beyond A1–A3, BI parity, and classical group-classification results. As a corollary, the Heisenberg uncertainty principle is derived as a structural property of the non-resolved proto-state, not as an independent postulate (**proved**).

*Role:* Detailed proof of the algebraic core; cited by Q1 [22], Q2 [46].

### 3.5 No External Dimensional Scale at the Level of Admissibility

The noscale paper [32] establishes a structural rigidity result (Level 1): any deformation of the admissibility constraints that introduces a dimensional parameter independent of  $c_\chi$  breaks at least one of BFS-consistent spectral scaling, structural non-injectivity (A2/ENI), or bounded admissibility closure (A3–A4). Dimensional analysis within the framework is therefore fully constrained by  $c_\chi$  alone (**proved**).

*Role:* Closes the naturalness question for dimensional parameters.

### 3.6 Branch I Synthesis: NIF-Note

The papers ENI [34], Foundation [6], HeisenbergStructure [33], and noscale [32] form the *non-injective foundations sub-programme*, whose synthesis paper is the presentation note **NIF-Note** [65] (Presentation Note 5). NIF-Note maps these four constituents into the single axiomatic chain

$$A1-A4 \implies \Pi \text{ non-injective, } S_\Pi > 0 \implies [X, \sigma(X)] = Z \neq 0 \implies \text{Heis}_3(\mathbb{Z}/q\mathbb{Z}) \implies V_\rho,$$

and records the formal liquidation of the white paper’s preliminary  $\chi$ /relaxation/iterated-projection vocabulary: the substrate is static, the admissibility constraint replaces relaxation throughout the corpus, and observable evolution is an induced ordering of admissible projections rather than a dynamics of  $\chi$ . NIF-Note identifies two open structural deliverables for the sub-programme: extension of the Born rule beyond the SU(2) sector (Q3) and the Level 2 scale determination of noscale (every emergent scale as a functional of  $c_\chi$  alone).

*Role:* Synthesis of Branch I. Provides the explicit input/output table of the axiomatic foundations and is the canonical entry point for downstream sub-programmes (Notes 1–4 of the spectral admissibility, emergent geometry, gauge structure, and spectral gravity tracks).

## 4 Branch II: Spectral Admissibility Sub-Programme (O-Series)

The O-series targets the derivation of the charged-lepton mass hierarchy from the spectral structure of the admissible fibre. The central quantity is the power-law exponent  $\delta_{\text{pair}}$  of the pair-level capacity observable  $\sigma_{\text{pair}}(n) = \sigma_c(n) \cdot \sigma_{q-c}(n)$  along the BFS cascade. The main transfer chain is

$$c_\chi \longrightarrow \delta_{\text{pair}} \longrightarrow \beta^* \approx 0.126 \longrightarrow m_e : m_\mu : m_\tau. \quad (1)$$

This chain is proved unconditionally with respect to the fibre structure of  $\Pi$  (O24 [72], Corollary 4.2). The numerical validation of (1) is the subject of O25 [58].

The presentation note **SpAdm-Note** [67] is the synthesis paper of this sub-programme: it maps the constituent papers below — the five precursor papers and the O-series O1–O32 — into a single account of the derivation chain (1), recording the status of every result as proved, structural, numerical, or open, and stating the two remaining open deliverables (the pointwise proof of hypothesis [H-color] and the extension of the numerical campaign to  $q = 401$ ). Their individual contributions follow.

### 4.1 Precursor Papers: SU(2) Admissibility Sub-Programme

Three papers precede the O-series proper and explore spectral admissibility on Cayley graphs of finite SU(2) subgroups.

**SpectralAdmissibility** [52] (v2.0) carries out the spectral admissibility programme on two parallel chains. For the SU(2) chain  $Q_8 \subset 2I \subset \text{PSL}(2, \mathbb{F}_q)$ , it derives the admissibility hierarchy  $A_n^{\text{max}} = c_\chi / \sqrt{\lambda_n}$  in the linearised regime and shows that the non-linear DBI correction preserves it. For the SU(3) chain  $\Delta(27) \subset \Sigma(168)$ , it establishes via character theory the hierarchy  $\lambda_3 = \lambda_{\bar{3}} = 28 < \lambda_6 = \lambda_{\bar{6}} = 42 < \lambda_7 = 48$  on  $\text{Cay}(\Sigma(168), C_4)$ , proving that the fundamental and anti-fundamental representations are co-admissible and strictly dominate all competitors. The algebraic mechanism is uniform:  $\chi_3(g) = 1$  for all  $g \in C_4$  plays the same structural role as  $\chi_{\rho_4} = 0$  on  $Q_8$  and the golden-ratio identity on  $2I$  (**proved**).

**SpectralCapacity** [53] introduces the capacity functional  $\mathcal{C}(G, S)$  and proves the binary-polyhedral maximality conjecture: the adjoint identity  $\hat{A}_{\text{adj}} = 2M - I$  (where  $M$  is the second-moment tensor of the axis distribution) shows that icosahedral isotropy forces  $\mu_{\text{adj}}(2I) = -\frac{1}{3}$ , while axis-plane confinement forces  $\mu_{\text{adj}}(\text{Dic}_n) \geq 0$  for all  $n \geq 3$ ; combined with a Plancherel uniform bound, this yields a strictly increasing capacity gap  $\Delta(\alpha) > 0$  for all  $\alpha > 0$  (**proved**).

**SpectralGramRigidity** [37] (v1.0.1) shows that the geometry of neutral generating sets on SU(2) is fully encoded in pairwise character values  $G_{st} = -\frac{1}{2}\chi_\rho(st)$  (**proved**).



**Three Particle Generations** [68] derives the three-generation structure from the ADE stratigraphic levels of binary Cayley graphs (**structural**).

## 4.2 SpectralRelaxation: Expander Graph Dynamics

The SpectralRelaxation paper [8] establishes  $\Lambda_{\text{proj}} \asymp \lambda_2$  for LPS expander families (**proved**), providing the entry point for the O-series dynamics.

## 4.3 O1–O8: LPS Phase — Cascade Exponent $\beta$

**O1** [43]: LPS saturation is a fixed-valence artefact; growing valence restores the pre-saturation window (**proved**). (Note: O2 was not written; the O3 derivation path was chosen instead.)

**O3** [41]: First quantitative SM connection. The phenomenological window  $\beta^* \in (0.09, 0.13)$  is derived from the charged-lepton mass ratios  $m_e : m_\mu : m_\tau$  (**structural**).

**O4** [57]: The structural upper bound  $\beta \leq 1$  follows from the BI flux constraint and the Cheeger isoperimetric inequality (**proved**).

**O5** [5]: Obstruction theorem. The  $\beta$  gap cannot be closed by any purely representation-theoretic or fixed-representation approach (**proved**).

**O6** [30]: Matrix-level confinement of the cascade exponent (**proved**).

**O7** [39]: Reformulates the target as a capacity exponent  $\delta \in [7.4, 10.6]$  on three-step path fingerprints (**structural**).

**O8** [1]: Establishes the geometric obstruction closing the LPS phase. Exponential shell growth  $|S_n| \sim p^n$  on LPS graphs compresses the pre-saturation window to  $O(\log q)$  BFS steps (**proved**).

## 4.4 O9–O15: Heisenberg Transition

The geometric obstruction identified in O8 motivates replacing LPS graphs by Cayley graphs of  $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ .

**O9** [40]: Structural transition. Polynomial ball growth  $|B_n| \sim n^4$  provides  $\Theta(q)$  usable BFS steps (**proved**). This is the key structural transition of the O-series.

**O10** [38]: First large- $q$  computation ( $q = 101$ ). Confirms  $n^4$  growth; a TensorSketch representation obstruction is identified at large  $q$  (**proved**).

**O11** [73]: Bidimensional Fourier-character proxy overcomes the TensorSketch obstruction (**structural**).

**O12** [17]: Exact Weil-block fingerprints resolve the final obstruction. Establishes  $\hat{\delta}_{\text{exact}} \approx 4.3$ – $4.8$  for  $q \in \{29, 61, 101, 151, 211\}$  (**proved**).

**O13** [9]:  $\hat{\delta}_{\text{exact}}(q)$  decreases monotonically from 4.80 to 3.59; the finite-size hypothesis is definitively refuted (**proved**).

**O14** [36]: Corrects the  $\delta \mapsto \beta^*$  relation via block normalisation and central-phase contributions (**structural**).

**O15** [35]: Block-level no-go. No reweighting of individual block data can raise  $\hat{\delta}_{\text{exact}}$  to  $\delta \in [7.4, 10.6]$  (Corollary 4.8, **proved**). This result forces the pair observable introduced in O16.

## 4.5 O16–O24: Pair Observable and Unconditional Transfer Chain

**O16** [18]: Identifies the misidentification of the observable. The physically relevant quantity is  $\sigma_{\text{pair}}(n) = \sigma_c(n) \cdot \sigma_{q-c}(n)$  on conjugate pairs  $\{c, q-c\}$ . The exponent doubling  $\delta_{\text{pair}} = 2\delta_c \approx 7.44$  resolves the gap (**structural**).

**O17** [19]: Structural derivation.  $\rho_{q-c} = \bar{\rho}_c$  is forced by fibre-level admissibility; the pair structure is derived, not postulated (**proved**).

**O18** [31]: Foundational derivation of the parity involution  $c \leftrightarrow q-c$ . BI parity is the only global symmetry of  $S_{\text{BI}}[\chi]$ , ensuring each fibre of  $\Pi$  is exactly  $\{\chi, -\chi\}$  (minimal fibre condition, **proved**, Theorem 4.1).

**O19** [4]: The residual amplitude factor  $r(c, q) \in \{1, 2, 3, 4\}$  does not affect  $\delta_{\text{pair}}$ ; the observable is pipeline-independent (**proved**).

**O20** [42]: Dynamical selection criterion for admissible modes via projective persistence (**structural**).

**O21** [51]: Defines the observable saturation rank  $n_3^{\text{obs}}$  via  $\Sigma_c(n_3) = 3$  and constructs the transfer  $\delta_{\text{pair}} \rightarrow \beta^*$  (**structural**).

**O22** [50]: BI projection locking forces  $n_3^{\text{obs}}$  onto a BFS shell (Theorem 3.2, **proved**).

**O23** [69]: The threshold  $\Sigma_c(n_3) = 3$  follows from the quaternionic maximality of the admissible neutral generating set (**proved**).

**O24** [72]: Capstone of the analytical O-series. The Verticality Lemma (“ker  $\Pi$  is free, Im  $\Pi$  is constrained”) closes the unconditional transfer chain  $c_\chi \rightarrow \delta_{\text{pair}} \rightarrow \beta^*$  (Corollary 4.2:  $\beta^* \approx 0.126$ , **proved**).

#### 4.6 O25–O30, PTO, and Fibre Erasure: Numerical Validation, Sector Identification, Eigenvalue Ratio, and Information-Theoretic Axes

**O25** [58]: Systematic pair-level numerical campaign. Computes  $\delta_{\text{pair}}$  across all  $(q - 1)/2$  conjugate pairs with  $M = 50$  independent block samples per pair, for  $q \in \{29, 61, 101, 151, 211\}$ . Results:  $\bar{\delta}_{\text{pair}} \in \{8.89, 9.98, 9.55, 9.13, 8.78\}$  with inter-pair standard deviation decreasing from 0.54 to 0.11, confirming  $\delta_{\text{pair}}$  as a structural invariant. After the O14 normalisation correction, corrected values  $\delta_{\text{corr}}(q) \in [7.7, 9.0]$  fall consistently within the admissible window  $[7.4, 10.6]$  for  $q \in \{61, 101, 151, 211\}$ . Direct extrapolation to  $q \rightarrow \infty$  is shown to be structurally degenerate; the central open direction is the analytical control of  $n_1(q)/q$  (**structural**).

**O26** [45]: Dictionary between conjugate Weil blocks and rank-1 matrices in  $M_2(\mathbb{C})$ , connecting the Heisenberg fibre to  $2I \subset \text{SU}(2)$  (**structural**).

**O27** [47]: Every admissible morphism  $\Phi_{q,\rho}$  satisfying involution equivariance, quadratic compatibility, and naturality factors through  $\mathfrak{su}(2)$  (**proved**).

**O28** [7]: Two measurements from the Q5a-O5 checkpoints. Part A: the BFS window ratio  $n_1(q)/q$  decreases from 0.138 to 0.066 over  $q \in \{29, 61, 101, 151, 211\}$  (Scenario S2, pre-asymptotic regime); OLS fit  $n_1(q) = \hat{\alpha}q + \hat{\beta}$  gives  $\hat{\alpha} \approx 0.053$ ;  $\delta_{\text{corr}}(q) \in [7.4, 10.6]$  for all tested primes, confirming O25. Part B: formal effective-dimension computation of O26 Criterion 5.4 in  $\text{End}(H_{\text{eff}})$ ,  $H_{\text{eff}} = \mathbb{C}^3$  (HEFF\_DIM = 3, consistent with  $\Sigma_c(n_3) = 3$  from O23). The covariance  $\mathcal{C}_c$  has rank exactly 3 for every conjugate pair and every  $q \in \{61, 101, 151\}$  (100% of pairs), with invariant eigenvalue structure  $[\lambda_1 : \lambda_2 : \lambda_3] = [1 : \frac{1}{2} : \frac{1}{2}]$ . The exact degeneracy  $\lambda_2 = \lambda_3$  is a structural consequence of the Born–Infeld parity involution  $c \leftrightarrow q - c$  (O18), which correlates conjugate projections and constrains  $\mathcal{C}_c$  to a symmetric subspace. The gap from  $d_\rho^2 = 4$  reflects  $H_{\text{eff}} \neq V_\rho$ ; analysis of  $V_\rho \subset H_{\text{eff}}$  is carried out in O29.

**O29** [55]: Spin- $\frac{1}{2}$  sector identification via the symmetric rank formula (**structural**, numerically confirmed). The Born–Infeld parity constraint  $\pi_{q-c}(v) = \overline{\pi_c(v)}$  (O18) is proved to force every outer product  $M_j = \pi_c(v_j) \otimes \pi_{q-c}(v_j)^*$  into the complex symmetric subspace  $\text{Sym}(V_\rho, \mathbb{C}) \subset \text{End}(H_{\text{eff}})$ . This is an algebraic identity, not lifted by any complex-linear reparametrisation. The resulting covariance rank satisfies the *symmetric rank formula*  $r_{\text{eff}} = d_\rho(d_\rho + 1)/2$ , whose unique positive integer inversion at  $r_{\text{eff}} = 3$  gives  $d_\rho = 2$ , confirming the spin- $\frac{1}{2}$  sector. Numerically:  $r_{\text{eff}} = 3$  universally for  $q = 61$  (30/30 pairs) and  $q = 151$  (75/75 pairs) with zero inter-pair variance; at  $q = 211$  (105/105 pairs, 100%, multi-sample confirmed),  $r_{\text{eff}} = 3$  holds universally (the 2 single-sample anomalies confirmed as artefacts); at  $q = 101$ , modal  $r_{\text{eff}} = 3$  is confirmed by a dedicated multi-sample analysis ( $M = 5$  resamples per anomalous pair, 76.7% of (pair, sample) cells satisfy  $r_{\text{eff}} = 3$ , with modal  $r_{\text{eff}} = 3$  for all 18 pairs), confirming  $d_\rho = 2$  at the modal level. The O26 Criterion 5.4 is reformulated for conjugate-pair data:  $d_\rho^2 = 4 \rightarrow d_\rho(d_\rho + 1)/2$ ; spin- $\frac{1}{2}$  satisfies the reformulated criterion exactly.

**O30** [76]: Analytical derivation of the eigenvalue ratio  $\lambda_1/\lambda_2 = 2$  (**proved**). The Born–Infeld parity anti-linearity  $\rho_{q-c} = \bar{\rho}_c$  forces every admissible trajectory vector  $w_j = \pi_c(v_j) \in V_\rho \cong \mathbb{C}^2$



to satisfy  $|\alpha_j| = |\beta_j|$  (equatorial constraint), i.e.,  $w_j$  lives on an equatorial  $S^1 \subset S^3$  in  $V_\rho$ . In the equatorial regime, the central component  $\sqrt{2}\alpha_j\beta_j$  of  $\text{vec}(M_j)$  is phase-independent (the phases cancel exactly), while the diagonal components  $\alpha_j^2$  and  $\beta_j^2$  oscillate as  $e^{\pm 2i\phi_j}$ . The covariance then diagonalises under the single-harmonic phase decorrelation condition  $\langle e^{2i\phi_j} \rangle = 0$ , yielding  $\mathcal{C}_c = \langle r_j^4 \rangle \cdot \text{diag}(1/4, 1/2, 1/4)$  and spectral ratio  $[1 : \frac{1}{2} : \frac{1}{2}]$  exactly. The factor of 2 is purely structural: it is the squared symmetric-tensor normalisation coefficient in  $e_0 = (v_+ \otimes v_- + v_- \otimes v_+)/\sqrt{2}$ , isolated by the equatorial constraint as the sole source of eigenvalue asymmetry. The correction to the equatorial condition is  $O(q^{-1/2})$  from U1 [90]. This closes the open direction of O28–O29 §7.4.

**O31 [85]:** Structural framework for the SU(3) identification (**structural**). Shows the O23 Hurwitz argument cannot directly produce  $d_\rho = 3$  (proved). Defines the *colour triplet*  $\{c_1, c_2, c_3\} \subset (\mathbb{Z}/q\mathbb{Z})^*$  with  $c_1 + c_2 + c_3 \equiv 0 \pmod{q}$ ; such triplets exist iff  $q \equiv 1 \pmod{3}$  (proved). The metaplectic automorphism  $\phi_\omega(a, b, z) = (\omega a, \omega^{-1}b, z)$  forces  $\sigma_c(n) = \sigma_{\omega c}(n)$  for all shells on the colour-adapted Cayley graph  $\text{Cay}(G_q, \mathcal{S}_q^{(3)})$  (proved, Theorem 4.4). The symmetry group of the co-admissible triplet is exactly SU(3), with U(3), SO(3), U(2),  $G_2$  excluded (proved). The Standard Model gauge group  $G_{\text{SM}} = \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$  follows as a composite of three admissibility fixed points. In v1.5 the identification is made *unconditional* for the standard graph: each Born–Infeld fingerprint vector is a single additive character of frequency  $B = c_1b_1 + c_2b_2 + c_3b_3$  (Lemma single-frequency, proved), so the capacity observable  $\sigma_c(n)$  counts distinct frequencies and the orbit dilation  $B \mapsto \omega B$  preserves it exactly (Proposition hcolor-frequency, proved). This establishes [H-color] pointwise on the standard graph at finite  $q$ , so SU(3) and  $G_{\text{SM}}$  hold unconditionally for the standard graph. U1 universality implies  $|\sigma_c - \sigma_{\omega c}| = O(q^{-1/2})$  for any generating set, so the identification is generating-set-independent in the large- $q$  limit. The generating-set independence operates at two levels (*two-tier result*): (i) exactly, for the colour-adapted graph  $\text{Cay}(G_q, \mathcal{S}_q^{(3)})$  (metaplectic argument, Theorem 4.4); (ii) up to  $O(q^{-1/2})$ , for *any* generating set including  $\mathcal{S}_q$  (U1 universality, Proposition 4.7). Proposition 4.6 shows that  $\mathcal{S}_q^{(3)}$  is the *canonical* colour-sector generating set: the minimal extension of  $\mathcal{S}_q$  closed under  $\phi_\omega$ , in exact analogy with  $\mathcal{S}_q$  being canonical for the pair sector under  $\phi_{-1}$ . Additionally, Theorem 5.3 (*generator-twist isospectrality*) proves that the Weil-sector Markov operator is isospectral under the generator twist  $\mathcal{S}_q \mapsto \phi_\omega(\mathcal{S}_q)$ :  $\text{spec}(\rho_c(P_{\mathcal{S}_q})) = \text{spec}(\rho_c(P_{\phi_\omega(\mathcal{S}_q)}))$  for all  $c \in (\mathbb{Z}/q\mathbb{Z})^*$ , confirmed to machine precision ( $|\Delta\lambda| < 10^{-12}$ ) for all  $q \equiv 1 \pmod{3}$ ,  $q \leq 157$ . However, this isospectrality does *not* extend across distinct central characters:  $\text{spec}(\rho_c(P_{\mathcal{S}_q})) \neq \text{spec}(\rho_{\omega c}(P_{\mathcal{S}_q}))$  in general ( $\max |\Delta\lambda| \approx 0.3\text{--}0.4$ ), establishing that [H-color] on the standard graph cannot be proved through global Markov-spectrum equality; the BFS capacity is a strictly finer observable. The correct algebraic mechanism is identified by two new results. *Intra-coset Vandermonde symmetry* (Lemma, proved): the irreducibility of the Weil representation (Schur’s lemma) forces the Heisenberg centre to act as a scalar in each  $\rho_{c_i}$ , so every horizontal coset  $(a, b)$  contributes exactly one independent direction to each colour summand  $V_{c_i}$ ; the Vandermonde invertibility of the triplet phase matrix  $(\lambda_j^k)_{j,k}$ , with  $\lambda_j = e^{2\pi i c_j/q}$ , then guarantees this, independently of any spectral property of  $\rho_c(P_{\mathcal{S}_q})$ . *Colour sector rank equality* (Proposition, proved unconditionally):  $r_c(n) = r_{\omega c}(n) = r_{\omega^2 c}(n)$  for all BFS depths  $n$  and any seed, unconditionally on the standard graph. This proves [H-color] at the level of BFS sector ranks without reference to the global Markov spectrum; v1.5 strengthens this to the full O12/O32 capacity functional  $\sigma_c(n)$  via the single-frequency BI fingerprint mechanism above. The O32 numerical campaign provides independent confirmation across the four-prime dataset. Section 6 provides the conceptual unification: U(1), SU(2), SU(3) arise from BI-admissible fibre orbits of size 1, 2, 3 respectively (phase non-injectivity, minimal correlated non-injectivity, relational correlated non-injectivity), with non-factorisability (Foundation M, Axiom A3) as the uniform structural origin. **O32 [77]:** tests [H-color] numerically on  $q \in \{61, 151, 211, 307\}$  (test,  $q \equiv 1 \pmod{3}$ ) and  $q \in \{29, 101\}$  (controls). Noise floor **ctrl\_ref** =  $1.657 \times 10^{-3}$ . Normalised ratios:  $R = 2.1$  ( $q = 61$ , 1/4 triplets valid),  $R = 0.4$  ( $q = 151$ ),  $R = 1.0$  ( $q = 211$ ),  $R = 0.7$  ( $q = 307$ ). All four pass  $R \leq 3$ ;  $\delta_{\text{tri}} \approx 3\delta_c$  sub-percent;  $C_{\text{color}}$  numerical rank 1 in all cases. Additionally

proves analytically (Proposition avg-hcolor) that  $\mathbb{E}[\sigma_c(n)] = \mathbb{E}[\sigma_{\omega c}(n)] + O(q^{-1})$ , establishing  $[H\text{-color}]_{\text{eff}}$  (capacity-exponent equality) in the  $q \rightarrow \infty$  limit. In v1.5, the companion result of O31 (single-frequency BI fingerprint) makes pointwise  $[H\text{-color}]$  *exact* on the standard graph at finite  $q$ : the residual  $R_{\text{var}}$  is reinterpreted as the finite-sample footprint of independent auxiliary draws, with expected  $R_{\text{var}} \propto q^{-1}$  scaling consistent with the observed factor 5 decrease from  $q = 61$  to  $q = 151$ . Status: **structural** ( $[H\text{-color}]$  numerically confirmed for all  $q \in \{61, 151, 211, 307\}$ ;  $[H\text{-color}]_{\text{eff}}$  analytically established; pointwise  $[H\text{-color}]$  proved analytically via O31 v1.5).

**PTO** [44]: The cumulative projected capacity  $I(n) = \sigma_{\text{pair}}(0) - \sigma_{\text{pair}}(n)$  is strictly non-decreasing and hence compatible with an arrow of time. Confirmed numerically for  $q \in \{29, 61, 101, 151, 211\}$  (**structural**). The admissible cascade is shown to carry the structure of an oriented state machine: nodes are admissible projections, edges are admissible transitions, and branching weights play the role of quantum probabilities. The first transition  $U_0 \rightarrow U_1$  is guaranteed by projective incompleteness (Foundation Corollary to Theorem 5.4): temporal ordering starts as a structural necessity, not a postulate. Open: whether the characteristic shell  $n^*(q)$  coincides with the locking depth  $n_c(q)$  at which BFS coverage transitions from locally to universally dominated — which would connect the spectral ordering to the effective geometry results of Q10 [75].

**Fibre Erasure** [80]: Hub paper between ENI [34] and the BFS spectral cascade. Defines a residual fibre-information functional  $I(c; \sigma(\ell))$  on the BFS coarse-graining axis  $\ell$ , distinct from and orthogonal to the temporal axis  $n$  of PTO. Conditional on the structural sufficiency hypothesis  $[H\text{-suff}]$ , the data-processing inequality implies that  $I(c; \sigma(\ell))$  is non-increasing in  $\ell$ : fibre information is erased, not created, under admissible coarse-graining (Thm 4.1).  $[H\text{-suff}]$  is proved unconditionally at the BFS rank level (Prop. 3.6, label-blindness from Schur’s lemma); for the full Born–Infeld capacity it is numerically supported for  $q \in \{61, 151, 211\}$ . Strictly not a  $c$ -theorem: no counting of effective degrees of freedom is claimed (listed as open problem [O-2]). Together with PTO, constitutes the information-theoretic architecture of the admissibility cascade: resolution axis  $\ell$  (Fibre Erasure) and temporal axis  $n$  (PTO) are structurally orthogonal.

## 5 Branch III: Q-Series and Physical Applications

### 5.1 Q-Series: Quantum Mechanics and Spacetime Emergence

The Q-series derives quantum-mechanical and geometric structure from the Foundation axioms, taking as input the fibre identification  $F_n \cong V_\rho$  and selected O-series results.

**Q1** [22]: Phase coherence is a structural consequence of projective admissibility (not a postulate). Theorem 2.7 proves that any admissible transition preserves the BI indiscernibility of conjugate Weil blocks  $\rho_c$  and  $\rho_{q-c}$ , maintaining the metaplectic phase coherence of the fibre. The singlet correlator  $E(\hat{a}, \hat{b}) = -\hat{a} \cdot \hat{b}$  and the Tsirelson bound are derived from admissibility, O18, and O23 (**proved**; numerically validated for  $q = 29$ ).

**Q2** [46]: Extends the programme to spin- $\frac{3}{2}$ . On the binary icosahedral group  $2I$ , the spin- $\frac{1}{2}$  and spin- $\frac{3}{2}$  sectors share the Laplacian eigenvalue  $\lambda_{1/2} = \lambda_{3/2} = 18$  (co-admissibility). The Born rule, singlet correlator, and Tsirelson bound are derived for spin- $\frac{3}{2}$  without new postulates.  $SU(2)$  is established as the stable fixed point of the admissibility flow (**proved** for spin- $\frac{3}{2}$ ; general- $j$  extension proved in Q3 [91]).

**Q3** [91]: Completes the  $SU(2)$  quantum sector for all admissible  $j$ . BI indiscernibility forces the bipartite proto-state to be invariant under the diagonal  $2I$ -action on  $V_{\chi_{2j+1}} \otimes V_{\chi_{2j+1}}$ ; Schur’s lemma applied to the Clebsch–Gordan decomposition  $\chi_{2j+1} \otimes \chi_{2j+1} = \bigoplus_{J=0}^{2j} \chi_{2J+1}$  uniquely identifies the proto-state as the singlet  $|\Omega_j\rangle$ . The universal correlator  $E(\hat{a}, \hat{b}) = -j(j+1)/3 \cdot (\hat{a} \cdot \hat{b})$  and the Born rule are derived for all five admissible sectors  $j \in \{\frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}\}$  of  $2I$  (**proved**). Closes the open step of Q2 (identification of the isotropic fibre state with the proto-state for spin- $\frac{3}{2}$ , and general- $j$  extension).

The papers Q1, Q2, Q3, and the Bell paper [10] form the *quantum structure sub-programme*, whose synthesis paper is the presentation note **QS-Note** [66] (Presentation Note 7). QS-Note

organises these four constituents into the six-step structural chain *BI indiscernibility*  $\Rightarrow$  *phase coherence* (Q1 Thm 2.7)  $\Rightarrow$  *singlet correlator*  $E(\hat{a}, \hat{b}) = -\hat{a} \cdot \hat{b}$  (Q1 Thm 2.14, from four structural inputs)  $\Rightarrow$  *Tsirelson bound*  $|S_{\text{CHSH}}| \leq 2\sqrt{2}$  (Q1 Cor 2.16)  $\Rightarrow$  *Born rule* (Q1 Thm 2.18, no Gleason theorem)  $\Rightarrow$  *SU(2) as stable fixed point of the admissibility flow* (Q2 Thm 8.6)  $\Rightarrow$  *universal spin- $j$  correlator* (Q3 Thm 5.2 for all five admissible sectors of  $2I$ ). Bell [10] contributes the structural non-applicability of Bell-type factorizability in any non-injective effective description (published in *Quantum Reports*). The note isolates the upstream inputs consumed (Notes 1 and 5 for the Weil fibre and the admissibility thread; O18, O22, O23, Born-Infeld for the parity/isotropy/indiscernibility inputs; ENI for the Bell argument), records the proved/numerical/open status of every result, and states the two open deliverables: Born rule for general observables beyond SU(2), and numerical validation of the rank- $W_n^{(c)} = 0$  coherence witness across  $q \in \{29, 61, 101, 151, 211, 307, 401\}$ .

The papers Q5a, Q5a-O2, Q5b, Q6b, Q7–Q11, U1, W1, and H2 form the *emergent geometry sub-programme*, whose synthesis paper is the presentation note **EG-Note** [60]. It maps these twelve papers into the chain  $\Pi_q \rightarrow L_{\text{eff}} \rightarrow g^{\mu\nu} = 2\eta^{\mu\nu}$ , deriving the effective four-dimensional Lorentzian metric from the admissibility filter acting on the Weil representation, with all four metric coefficients fixed by the  $\mathfrak{su}(2)$ -Casimir on the spin-1 module  $\text{Sym}^2(V_\rho)$ . Their individual contributions follow.

**Q5a** [26]: Under Mosco hypotheses, the admissibility forms  $\mathcal{E}_q$  on  $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$  converge to  $L_\Pi = -A\partial_x^2$  on  $L^2(\mathbb{R})$  (**structural**, conditional).

**Q5b** [11]: The BFS shell stratification of  $\text{Heis}_3(\mathbb{R})$  produces a four-dimensional Lorentzian spacetime, with the temporal direction identified with the BFS depth. Hypothesis [H-lift] (Q5b-O1), formerly conditional, is proved by Q9 [81] (**structural**; unconditional with respect to [H-lift] via Q9).

**Q6a** [23]: Gauge charges are the invariants of the projection-fibre equivalence classes. U(1) and SU(2) are identified as fixed points of the admissibility flow.  $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$  is derived as a composite (**structural**; SU(3): unconditional for the standard graph via the v1.5 single-frequency BI fingerprint mechanism, see O31 [85]).

**Q6b** [16]: Effective Lorentzian geometry from the admissibility filter  $\Pi_q$ ; inherits Q5a hypotheses only ([H-lift] closed by Q9). Derives the effective metric  $g^{\mu\nu}(x) \propto A^{\mu\nu}(x)$  of Lorentzian signature  $(-, +, +, +)$  from the principal symbol of  $\mathcal{L}_{\text{eff}}$ ; with Q8, Q10, Q11 the co-metric is fully explicit:  $g^{\mu\nu} = \text{diag}(-2, 2, 2, 2)$ . Proves the Schwarzschild metric as the unique stationary exterior solution under spherical symmetry and flux conservation; the horizon appears as a principal-symbol degeneracy. Establishes the Hamilton–Jacobi equation  $g^{\mu\nu} \partial_\mu S \partial_\nu S = 0$  as the effective massless propagation structure. The Einstein equations emerge as consistency conditions via the spectral entropy functional. Closes the geometric chain  $\Pi_q \xrightarrow{\text{Q5a}} \mathcal{L}_\Pi \xrightarrow{\text{Q5b}} g^{\mu\nu} \xrightarrow{\text{Gravity}} G_{\mu\nu}$  (**structural**, conditional on Q5a).

**Q7** [89]: Bridge candidate between the admissible dimension  $\dim H_{\text{eff}} = 3$  (O23–O29) and the three spatial dimensions of Q5b. Proves  $[\mathfrak{su}(2)] \not\cong \mathfrak{heis}_3$  (the bridge cannot be a Lie algebra isomorphism) and identifies  $H_{\text{eff}} \cong \text{Sym}^2(V_\rho)$  as the spin-1 representation of  $\mathfrak{su}(2)$  (O29). Establishes structural uniqueness of the bridge via Schur’s lemma, and reduces the identification to the isotropy condition  $A_H = A_z$  on  $\sigma_2(L_{\text{eff}})$ . Proves analytically  $[F_c, L_{\text{Weil}}] = 0$ , which forces the cross terms  $k_X k_Z = k_Y k_Z = 0$  structurally. Numerical evidence on O25 checkpoints ( $q \in \{61, 101, 151, 211\}$ ) gives  $A_z = 2$  universally and  $|A_H - A_z| \rightarrow 0$ ; four-point OLS fit gives  $|A_H - A_z| \approx 0.364 q^{-0.52}$  ( $R^2 = 0.98$ ), consistent with  $O(q^{-1/2})$  (Q5b Theorem 3.2). (**structural**; asymptotic isotropy  $A_H \rightarrow 2 = A_Z$  proved by Q10 and U1, conditionally on [U]; bridge existence at finite  $q$ : **open**).

**Q8–Q11, U1, W1, H2 (Geometric emergence cluster)**. Q8 [56] establishes  $A_Z = C_{\mathfrak{su}(2)} = 2$  unconditionally under Q5a via Casimir rigidity (**structural**). Q9 [81] proves [H-lift] as a theorem: the modulation generator is suppressed at rate  $O(q^{-1/2})$ , and  $A_H > 0$  follows by coercivity independently of the bridge (**proved**, under Q5a). Q10 [75] proves  $A_H \rightarrow 2$  under spectral

universality [U]: character-independence forces  $\mathfrak{su}(2)$ -isotropy of the effective form, and the unique such form is the Casimir with value 2 (**structural**, conditional on [U]). U1 [90] proves [U] with rate  $\varepsilon(q) = O(q^{-1/2})$  from the Weil-generator Lipschitz bound and BFS–Carnot–Carathéodory convergence (**proved**). W1 [92] proves Hypothesis [H-w] (convergence  $a_q(s) \rightarrow A > 0$  of the admissibility weights) as a structural consequence of the fibre-invariance of admissible observables ( $\mathcal{O} = \text{Im } \Pi$ ), made quantitative by U1 (**proved**). After W1, the open Q5a hypotheses reduce to three: [H1], [H-E1], and [C]. Q5a-O2 [84] subsequently proves that [H1] is not needed in full generality on the admissible sector, and closes [H-E1] and [C] via spectral atomicity. H2 [83] proves the last open hypothesis, [H2] (strong convergence of the rescaled Weil generators  $\hat{X}_q \rightarrow x$  and  $\hat{P}_q \rightarrow -i\partial_x$ ), via a quantitative Poisson aliasing lemma and a discrete Sobolev identity (**proved**). All hypotheses of the Q5a large- $q$  convergence theorem are verified unconditionally on the admissible sector. **Q11** [86]: Temporal Casimir rigidity closes Q5b open problem O3 (identification of  $A_\tau$  from admissibility data) (**structural**). The cascade increment operator internalises no new irreducible representation (Schur rigidity); asymptotic  $\mathfrak{su}(2)$ -equivariance forces the temporal coefficient  $A_\tau = 2$ , identical to the spatial Casimir value. The effective Lorentzian co-metric is thereby fully identified:  $g^{\mu\nu} = \text{diag}(-2, 2, 2, 2) \propto \eta^{\mu\nu}$  (Minkowski). The derivation chain  $\text{Q5a} \rightarrow \text{Q5b} \rightarrow \text{Q7} \rightarrow \text{Q8} \rightarrow \text{Q9} \rightarrow \text{Q10} \rightarrow \text{U1} \rightarrow \text{W1} \rightarrow \text{H2} \rightarrow \text{Q11} \rightarrow \text{Q5b-O3}$  closed is now complete.

**Q12** [93]: Yang–Mills dynamics from the vertical variation of the projective spectral entropy (**structural**, conditional on Q6a for the gauge group; SU(3) sector unconditional at the pointwise level:  $[H\text{-color}]_{\text{pointwise}}$  proved analytically in [85]). The Gravity paper [25] derived the Einstein tensor as the infrared-dominant horizontal response of  $S_\Pi[g] = \frac{1}{2} \log \det' A_g$  (the  $a_2$  coefficient). Q12 carries out the complementary vertical variation: the operator  $A_g$  is extended to  $A_{g,\mathcal{A}} = -(\nabla^{\mathcal{A}})^2 + E$  on the associated vector bundle of the admissible principal bundle  $P_{G_\Pi}(M, G_\Pi)$ , and the Seeley–DeWitt coefficient  $a_4$  is computed:

$$a_4(A_{g,\mathcal{A}})|_{\text{gauge}} = \frac{1}{16\pi^2} \int \frac{1}{12} \text{Tr}(F_{\mu\nu} F^{\mu\nu}) \sqrt{g} d^4x.$$

The extended spectral entropy  $S_\Pi[g, \mathcal{A}]$  therefore contains the renormalized Yang–Mills term with logarithmic coupling  $g_{\text{YM}}^{-2} \sim (12 \cdot 16\pi^2)^{-1} \log(\Lambda/\mu)$ , and the vertical variation  $\delta_{\mathcal{A}} S_\Pi = 0$  yields the Yang–Mills equations  $D_\mu F^{a\mu\nu} = 0$  in the current-free sector (**proved** for the  $a_4$  derivation given  $G_\Pi$ ). The induced Newton’s constant  $G_N^{-1} \sim \ell_{\text{sp}}^{-2}$  (quadratic UV) and the induced gauge coupling (logarithmic UV) share the same spectral cutoff  $\ell_{\text{sp}}$ ; the gauge–gravity hierarchy  $G_N g_{\text{YM}}^2 \ll 1$  is a structural consequence of the Seeley–DeWitt expansion and requires no fine-tuning. The horizontal–vertical decoupling lemma guarantees that the Einstein and Yang–Mills sectors are independent at orders  $a_2$  and  $a_4$ :  $a_2 \rightarrow \text{Einstein}$ ,  $a_4 \rightarrow \text{Yang–Mills}$ , from the single functional  $S_\Pi[g, \mathcal{A}]$ . Beyond the technical derivation, Q12 identifies a conceptual principle: gravity and gauge dynamics are separated by geometric direction (horizontal vs. vertical variation on the admissible bundle) *and* by spectral order in the Seeley–DeWitt hierarchy, not by a missing symmetry group. The distinct UV behaviours — quadratic divergence for gravity, logarithmic for gauge — are structural consequences of this spectral stratification and require no independent input. This suggests that the natural organisational space for unification in the present framework is spectral and variational rather than primarily group-theoretic: the difficulty of reconciling gravity with the gauge-theoretic Standard Model may reflect a difference of Seeley–DeWitt order rather than the absence of a common symmetry group  $G_{\text{unif}}$ . The gauge–gravity synthesis is carried out in Q13.

**Q13** [79]: Gauge–gravity spectral synthesis (**structural**, conditional on Q12 and the Gravity paper; SU(3) sector unconditional at the pointwise level ( $[H\text{-color}]_{\text{pointwise}}$  proved in [85])). Q13 completes the trilogy (Gravity, Q12, Q13) by solving the *joint* variational problem  $\delta_{g,\mathcal{A}} S_\Pi = 0$  and deriving the coupled Einstein–Yang–Mills system. The metric variation of the  $a_4$  coefficient yields the Yang–Mills stress tensor as the source term for gravity:  $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}^{\text{YM}} + \mathcal{O}(a_6)$  (**proved** modulo the Lemma on metric-independence of  $F_{\mu\nu}$ ). The  $a_6$  Seeley–DeWitt coefficient is computed and produces the leading gauge–gravity cross-coupling  $R_{\mu\nu\rho\sigma} F^{\mu\nu} F^{\rho\sigma}$ , suppressed by  $\ell_{\text{sp}}^2$



(**structural**). The Eddington–Born–Infeld completion of the joint functional is constructed from the same  $c_\chi$  saturation bound, recovering both sector-specific BI completions as limits (**structural**, with residual ambiguity in the tensor  $\Xi_{\mu\nu}$ ). The gauge–gravity hierarchy is a structural output:

$$G_N g_{\text{YM}}^2 \sim \frac{\ell_{\text{sp}}^2}{\dim V} \cdot \left[ \log \frac{\Lambda}{\mu} \right]^{-1} \ll 1,$$

without fine-tuning, as a direct consequence of the Seeley–DeWitt stratification ( $a_2 \rightarrow$  gravity at quadratic UV,  $a_4 \rightarrow$  gauge at logarithmic UV). The central conceptual contribution is the identification of the spectral stratification principle: gravity and gauge dynamics are not unified by enlarging the symmetry group, but are *stratified by Seeley–DeWitt order* within the single operator  $\mathcal{A}_{g,A}$ . The fermionic extension is carried out in Q14.

Q12 and Q13 form the *gauge–gravity stratification sub-programme*, whose synthesis paper is the presentation note **GGs-Note** [62] (Presentation Note 8). GGS-Note organises the two constituents into the spectral stratification chain  $a_2 \rightarrow G_{\mu\nu}$  (horizontal  $\delta_g$ ),  $a_4 \rightarrow D_\mu F^{a\mu\nu} = 0$  (vertical  $\delta_A$ ),  $a_6 \rightarrow R_{\mu\nu\rho\sigma} F^{\mu\nu} F^{\rho\sigma}$  (mixed, structural), all from the single functional  $S_\Pi[g, A] = \frac{1}{2} \log \det' \mathcal{A}_{g,A}$ . Q12 is shared with the gauge-structure sub-programme (Note 3), where it provides the gauge group identification; GGS-Note draws on its  $a_4$  vertical variation, on the horizontal–vertical decoupling Lemma 1, and on the UV-hierarchy comparison  $G_N^{-1} \sim \ell_{\text{sp}}^{-2}$  vs.  $g_{\text{YM}}^{-2} \sim \log(\Lambda/\mu)$ . Q13 supplies the coupled Einstein–Yang–Mills system, the  $a_6$  cross-coupling, the Eddington–Born–Infeld joint completion (conditional on [H-ext]), and the structural hierarchy  $G_N g_{\text{YM}}^2 \ll 1$ . The note isolates the upstream inputs consumed (Note 4 for  $S_\Pi[g]$  and the  $a_2$  sector; Note 3 for  $G_\Pi$  and the principal bundle; Note 2 for the effective Lorentzian base manifold; BornInfeld for the saturation bound  $c_\chi$  and the conditionality [H-ext]) and records the three open deliverables:  $a_6$  phenomenological normalisation, analytical derivation of [H-ext] from A1–A4, and the coupled equations with fermionic matter currents from Q14 (Note 6). Per O31 Prop. 4.23,  $[H\text{-color}]_{\text{pointwise}}$  is proved on the standard graph at finite  $g$ , making  $G_\Pi = \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$  an unconditional input to the stratification chain; the  $a_4$  Yang–Mills derivation of Q12 therefore holds unconditionally for the full Standard Model gauge group.

**Q14** [78]: Fermionic matter and chirality (**structural**, conditional on Q13, the Gravity paper, and the BI paper; colour sector unconditional at the pointwise level ( $[H\text{-color}]_{\text{pointwise}}$  proved in [85])). Q14 extends the gauge–gravity synthesis to the fermionic sector. Fermions are not postulated as external fields: they arise as the spinorial face of the Weil module  $V_\rho$  already forced by non-injective admissibility, once its metaplectic Lie-algebraic structure is complexified by the Lorentzian metric. Three structural results are proved. (i) The admissible Weil module induces, through its metaplectic lift and the complexification  $\mathfrak{mp}(2, \mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C})$ , an admissible spinor bundle  $S_\Pi$  whose symmetric and determinant tensor sectors reproduce the  $\text{SU}(2)_L \times \text{U}(1)_Y$  electroweak bundle structure (**structural**). (ii) The projected Dirac operator  $\mathcal{D}_{\Pi,g,A}$  contains a canonical zero-order endomorphism  $E_\Pi$  (the internal spectral residue of non-injective projection) which, via the spinorial lift of BI parity (proved in Q14 as the antilinear  $J_\Pi = HK$ ), is left-admissible and enforces the  $V$ – $A$  chiral structure; the  $\gamma_5$ -weighted  $a_4$  Seeley–DeWitt coefficient then imposes anomaly-cancellation trace constraints on the hypercharge weights, making hypercharge a spectral coherence datum rather than a free parameter (**structural**). (iii) The saturation invariant  $\sigma_c(n_3) = 3$  admits a spinorial multiplicity reading, distinct from its geometric reading as three spatial directions, yielding a gauge-singlet three-generation factor  $\mathbb{C}_{\text{gen}}^3 \subset \ker(\text{ad}_{\text{SU}(2)} \oplus Y)$  (**structural**). The colour-coupled quark sector follows by tensoring with  $V_{\text{color}}$ , unconditional at the pointwise level ( $[H\text{-color}]_{\text{pointwise}}$  proved in [85]). Beyond these three theorems, Q14 §6 establishes a qualitative mechanism for the inter-generation splitting (a diagnostic model, not a fourth theorem). Identifying the rank-three admissible sector with  $\text{Sym}^2(V)$  in the End reading of O29 [55] and O30 [76], a static  $J_\Pi$ -real, weight-preserving restriction of  $E_\Pi^2$  has a degenerate outer pair (static obstruction), while the oriented cascade generator  $\partial_\tau$  of Q11 [86] carries a non-zero  $J_3$  projection that lifts it. The qualitative mechanism is thereby fixed; its amplitude is deferred to the cascade normalisation, and generation mixing requires the complex metaplectic phase.

Q14 is the principal constituent of the *fermionic matter sub-programme*, whose synthesis paper is the presentation note **FM-Note** [61] (Presentation Note 6). FM-Note organises Theorems A–C into the chain  $\Pi \Rightarrow F_n \simeq V_\rho \Rightarrow \text{Mp}(2, \mathbb{R}) \Rightarrow \text{mp}(2, \mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C}) \rightsquigarrow \text{Spin}(3, 1) \simeq \text{SL}(2, \mathbb{C}) \Rightarrow \mathcal{S}_\Pi$ , isolates the four upstream layers consumed (Notes 1, 5 for the Weil fibre; Note 2 for the Lorentzian closure; Note 3 for the gauge bundle; Q13 [79] for the gauge–gravity synthesis), and records the two open deliverables of the sub-programme: explicit construction of  $E_\Pi$  as a spectral operator, and the full Lorentzian-signature matter content  $\mathcal{S}_\Pi^{\text{matter}} = (\mathcal{S}_\Pi \oplus (\mathcal{S}_\Pi \otimes V_{\text{color}})) \otimes \mathbb{C}_{\text{gen}}^3$  with explicit Yukawa structure. The qualitative mechanism of the generation splitting is now fixed (Q14 §6); its amplitude, via the cascade normalisation, is the remaining quantitative step.

**PRS** [88]: The projective residue as a Schur complement (**structural**). Companion note of the fermionic matter sub-programme refining the Q14 §6 open deliverable on the inter-generation splitting. Exhibits  $E_\Pi = -\Pi_S D(1-P) D \Pi_S^* = -M^\dagger M$  as the Schur complement of the spinorial directions eliminated by the non-injective projection (negative semi-definite, zero-order, vanishing in the injective limit), localises the split coefficient  $u$  in the  $\mathcal{D}^\pm$ -transported, generation-projected component of the antiunitary chiral equivariance defect  $\Delta_\chi(P) = \pi_{LL} - \tau \overline{\pi_{RR}} \tau^{-1}$ , and stratifies the result on the Cosmochrony axioms: A1–A3 force  $u = 0$  (chirally symmetric minimal non-injectivity); a non-zero  $u$  requires chiral symmetry-breaking carried by the projection-locking axiom A4. Establishes the finite/Lorentzian separation theorem ( $u_{\text{fin}} = 0$  for  $q$  an odd prime), locks the Seeley–DeWitt conventions so that the operator-level and spectral-level  $u$  coincide on the flat effective metric, and provides a no-go result: chirality is Lorentzian, not a finite-fibre label. Reduces the remaining open problem to the Lorentzian chiral curvature  $\mu_\chi^2 := \partial_s^2 \mathcal{B}(0)$  of the saturated Born–Infeld functional along the  $J_\Pi$ -odd modulus.

**AAR** [82]: Reduction of the angular amplitude observable (**structural**, working note). Fixes, before any Weil–BFS computation, the precise observable whose measurement tests the absolute normalisation of the Q14 §6 angular generation split. A short Baker–Campbell–Hausdorff reduction identifies the per-step  $J_\Pi$ -odd  $J_3$  component of the metaplectic cascade generator with the oriented symplectic area swept per step — the central Carnot-degree-2 increment of  $\text{Heis}_3$  — not a free shear parameter. The quantity to be measured is the accumulated oriented  $J_3$ -odd area along the Weil–BFS cascade, normalised by the projected capacity  $\hat{I}(n)$ ; at  $n_3^{\text{obs}}$  this equals  $\varepsilon_{\text{Weil}}(n_3^{\text{obs}})$ , to be compared with the dictionary value  $\varepsilon = 1/10$  without fitting. In the O12 fingerprint the two sectors are the two terms of the exponent: frequency  $B_c$  carries the radial capacity  $\sigma_c$ , central phase  $A_c$  carries the angular split. A direct evaluation on symmetric BFS shells returns  $\Theta_{\text{raw}} = 0$  identically; the orientation prescription is an irreducible input prior to  $\mathcal{N}_A$ .

**Q11OF** [87]: The oriented frontier observable (**structural**, diagnostic). Frontier-transfer diagnostic note on the recursive existence of the  $J_3$ -odd cascade signal underlying the Q14 §6 splitting mechanism. The symmetric-capacity obstruction makes the raw shell observable vanish identically ( $\Theta_{\text{raw}} = 0$ ): the involution  $g \mapsto g^{-1}$  pairs opposite central phases on symmetric BFS shells, erasing the oriented  $J_\Pi$ -odd component before the radial recursion can act. The honest replacement is a frontier transfer observable on the directed outgoing edges  $g \rightarrow gs$ , whose central increment is the Heisenberg cocycle  $a(g) s_b$ , discrete counterpart of the BCH term  $\frac{1}{2}[X, Y]$  and hence of the dynamical  $J_3$ . The orientation is the recursive arrow  $\partial_\tau = \log g$  from the projection origin. The first test is the existence of an oriented signal  $\langle \Delta A_c \rangle_{\partial + S_m} \neq 0$ ; the discriminating anti-bias control is the symmetrised-frontier cancellation. No normalisation, no dictionary value, no  $\varepsilon$  enters: the note tests only whether the recursion produces a non-zero, bias-free oriented signal.

## 5.2 Gravity and Spacetime Track

**Spectral2** [48]: Derives operational metric distances from graph Laplacians on a relational substrate without any background manifold (**structural**).

**Gravity** [25]: Derives  $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}^{(\Pi)}$  from the metric variation of  $S_\Pi[g] = \frac{1}{2} \log \det' A_g$ , via heat-kernel and zeta regularisation (**proved** at leading IR order). Establishes that the UV comple-



tion is uniquely determined to be of determinantal Born–Infeld form  $\sqrt{-\det(g_{\mu\nu} + \ell_{\text{sp}}^2 R_{\mu\nu})} - \sqrt{-g}$ , with Einstein gravity as the universal two-derivative IR sector (**proved**).

**Thermodynamics** [28]: Develops a local thermodynamic interpretation. The diagonal heat kernel yields a local multiplier field  $\beta(x)^{-1} = \beta_0^{-1} - \frac{1}{6}R(x) + O(\nabla^2 R)$ , and a spectral first law  $\delta S_{\text{II}} = \int \beta(x)^{-1} \delta E_{\text{II}}(x)$  recasts the Einstein equation as a spectral equilibrium condition (**structural**; Lorentzian extension and matter coupling: **open**).

**Lorentz** [13]: Lorentzian completion of the spectral action. Causal propagation and gravitational wave propagation follow from the hyperbolic continuation (**proved** for the Lorentzian extension; GW spectrum: **structural**).

### 5.3 Lorentz Capacity Sub-Programme

The Lorentz-capacity (LC) sub-programme isolates a single structural question: where does the Lorentz factor  $\gamma$  originate, given that the Lorentzian metric is itself an effective object reconstructed from admissible non-injective projection? The answer is that  $\gamma$  is not a kinematic postulate but the temporal residual of a bounded Born–Infeld capacity budget, lifted to special-relativistic observer transformations through the Q5b–Q11 effective co-metric  $g^{\mu\nu} = 2\eta^{\mu\nu}$ . A guiding distinction runs throughout: the Born–Infeld capacity sphere  $(F^\tau)^2 + \beta^2 = 1$  supplies the load  $\beta$  and the clock factor  $1/\gamma$ , while the Lorentz group acts as the isometry group of the effective metric, not of the capacity sphere. The completed chain is

$$\text{BI budget} \rightarrow F^\tau = 1/\gamma \rightarrow g^{\mu\nu} = 2\eta^{\mu\nu} \rightarrow \Lambda(\beta) \rightarrow R(x)^2 = 1 - \epsilon(x) \rightarrow \text{finite-}q \text{ domain.}$$

The synthesis paper **LC-Synthesis** [64] gathers the six constituent papers below into this single account; their individual contributions follow.

**LorentzCapacity** [54]: Derives the Lorentz factor and relativistic mobility  $\beta\gamma$  from the O7 saturation parameter  $\eta_n^{\text{O7}}$  under the Born–Infeld admissibility budget. The analytic factorization (central coordinate  $c$  contributes only a global phase) shows the total O7 residual satisfies a linear filling law and does not by itself yield the Lorentz factor. The conditional mechanism via the temporally projected residual  $\Sigma_n^\tau := \mathcal{P}_\tau^{\text{Q5b}} \Sigma_n^{\text{tot}}$  and a quadratic normalization gives  $\eta_n^\tau \rightarrow \beta\gamma$  and  $\Phi(\eta_n^\tau) \rightarrow 1/\gamma$  (**proved** for the algebraic BI target and the factorization;  $\mathcal{P}_\tau^{\text{Q5b}}$  construction: closed by TempProj [59]).

**TempProj** [59]: Closes the open construction of  $\mathcal{P}_\tau^{\text{Q5b}}$  from LorentzCapacity. Constructs the temporal residual map as the unique positive function compatible with the Lorentzian splitting of the Q5b completed co-metric  $g^{\mu\nu} = 2\eta^{\mu\nu}$ , the Born–Infeld unit capacity-flow sphere, and the exact Weil filling law  $\Sigma_n^{\text{tot}} = 1 - B_n$ . The explicit formula  $\mathcal{P}_\tau^{\text{Q5b}} \Sigma_n^{\text{tot}} = \sqrt{1 - B_n^2}$  follows from a fibre-by-fibre uniqueness argument; Lemma 1 and Theorem 1 of LorentzCapacity become unconditional (**proved** conditional on [U]; three-input necessity: **proved**).

**LCII** [20]: Extends TempProj to the non-homogeneous fibrewise regime. A localised spectral occupancy  $\epsilon(x)$  reduces the local BI capacity radius to  $R(x)^2 = 1 - \epsilon(x)$ ; the fibrewise map factorises as  $\mathcal{P}_{\tau,x}^{\text{Q5b}} \Sigma^{\text{tot}}(x, n) = R(x) \sqrt{f(x, n)(2 - f(x, n))}$ . Gravitational time dilation is identified as  $d\tau/dn|_x = R(x)/\gamma$ , with  $R(x)^2 = -2g_{\tau\tau}(x)$  linking the local capacity budget to the metric. The paper establishes a unified capacity arbitration picture: special-relativistic time dilation ( $1/\gamma$ ) and gravitational time dilation ( $R(x)$ ) are two independent manifestations of the same BI capacity trade-off, separated in the product  $R(x)/\gamma$  (**structural**; open problem LC-O2-O1 closed by [27]).

**LCIIo1** [27]: Closes the capacity–metric bridge of LCII spectrally, without importing the Einstein equation. Defining the local occupancy as a quadratic norm on the rank-three sector  $\text{Sym}^2(V_\rho)$ , the  $\text{SU}(2)$ -equivariance of the admissibility weight (Schur’s lemma) forces the local temporal coefficient of  $L_{\text{eff}}$  to satisfy  $A_\tau(x) = 2(1 - \epsilon(x))$ . The principal-symbol identification [16, 86], applied locally with the flat-space normalisation  $A_\tau^\infty = 2$ ,  $g^{\tau\tau, \infty} = -\frac{1}{2}$ , yields  $g^{\tau\tau}(x) = -A_\tau(x)/4$ , hence  $R(x)^2 = 1 - \epsilon(x) = -2g^{\tau\tau}(x)$ . The open bridge of Remark 3.2 of [20] becomes a theorem; the chain  $\epsilon(x) \mapsto A_\tau(x) \mapsto g^{\tau\tau}(x)$  is closed spectrally (**proved**).

**LCo1** [29]: Closes LC-O1, the reconstruction of effective Lorentz transformations between inertial observers from projective temporal ordering data. Given a projective temporal capacity load  $\beta$  from LorCap/TempProj, the unique proper orthochronous transformation between the corresponding inertial splittings of the effective co-metric  $g^{\mu\nu} = 2\eta^{\mu\nu}$  is the standard Lorentz boost; length contraction  $\ell = \ell_0/\gamma$  follows as a corollary (**proved**).

**LCo3** [21]: Closes LC-O3, the finite- $q$  correction problem near the saturation boundary. Setting  $R := (q - 1)/2$  and  $n = R + m$ , the parabolic Carnot formula deviates from the exact torus capacity by  $\Delta_q(n) = 4m^2/q^2$ ; the quadratic identity  $B_n^2 + (F_n^\tau)^2 = 1$  receives no correction, while the Carnot identification holds for  $n \leq R$  and asymptotically as  $m/q \rightarrow 0$  (**proved**). With LC-O1, LC-O2-O1, and LC-O3 now closed, the Lorentz-capacity sub-programme is complete.

## 5.4 Standard Model Companion Papers

The papers Foundation, TopologicalInvariants, SMchargemass, O23, O27–O32, Q6a, Q7, and Q12 form (in their gauge-group identification aspect) the *gauge structure sub-programme*, whose synthesis paper is the presentation note **GS-Note** [63]. It identifies the admissible internal symmetry group  $G_\Pi = \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$  as a forced output of the admissibility structure of  $\Pi$ , organising the  $\text{U}(1)$  phase-fibre sector (Hopf fibration), the  $\text{SU}(2)$  quaternionic sector (O27 rigidity), and the  $\text{SU}(3)$  colour-triplet sector (O31–O32 co-admissibility). The note covers gauge group identification only; the derivation of Yang–Mills dynamics belongs to the gauge–gravity spectral stratification sub-programme.

**BornInfeld** [12]: BI action as the unique minimal local representation consistent with a universal bound on admissible relational gradients (**proved**).

**TopologicalInvariants** [70]: Charge quantisation ( $\pi_1(S^1) \cong \mathbb{Z}$ ) and absence of magnetic monopoles from the Hopf fibration  $S^1 \hookrightarrow S^3 \rightarrow S^2$  of the projection fibre (**proved**;  $\pi_2(\mathcal{C}_{\text{eff}}) = 0$  beyond the canonical model: **open**, Remark 4.2).

**SMchargemass** [14]: Mass and electric charge as BI-saturated responses (**structural**; baryon-to-photon ratio  $\eta$ : **open**).

**Bell** [10]: Bell-inequality violations from non-injectivity alone, without nonlocality or additional hidden variables (**proved**).

The Bell paper opens the *entanglement sub-programme*, whose presentation paper is **ENT-Note** [49] (v1.1). The note locates entanglement *inside* the spectral admissibility cascade: for a closed conjugate Weil pair  $\{c, q - c\}$ , the reduced entropy on the residual fibre-level support of the Gram–Schmidt span is exactly  $S_{\text{ent}}(n) = \log r_{\text{pair}}(n)$ , the logarithm of the residual Gram–Schmidt rank, with no surviving spectral weighting. Structural monotonicity of  $S_{\text{ent}}$  along the cascade follows from the Gram–Schmidt construction alone. v1.1 integrates the analytical results of ENT-E1 (Weil fingerprint orthogonality theorem) and ENT-E2 (separation of the admissible pair entropy  $S_{\text{ent}}^{\text{adm}} = \log 2$  from the full residual-rank entropy), upgrading several previously numerical hypotheses to theorems and recording the residual epistemic stratification. The note thus reduces entanglement to a pure O-series observable and connects the projected mass hierarchy — tracked by the pair-capacity exponent  $\delta_{\text{pair}}$  — to a downstream trace of the stabilization profile.

**ENT-E1** [74]: closes the open analytic strengthening of ENT-Note by explicit computation. The  $k=3$  Weil fingerprint of  $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$  for any block  $(c_1, c_2, c_3)$  factors as a global phase times a single discrete Fourier mode at frequency  $A = c_1b_1 + c_2b_2 + c_3b_3 \bmod q$ . Since distinct Fourier modes are orthogonal, every newly admitted fingerprint is already orthogonal to the current span, so the admission residual  $w_j = 1$  is an *exact* algebraic identity — not a numerical accident. For matched single-character blocks the conjugate pair samples sign-reflected frequencies and the closure hypothesis of ENT-Note becomes a theorem; the flat Schmidt spectrum  $S_{\text{ent}}(n) = \log r_{\text{pair}}(n)$  is then derived analytically (**proved**; literal Schmidt pairing for independently sampled canonical O25 blocks: **open**).

**ENT-E2** [71]: separates two genuinely distinct entanglement entropies that ENT-Note and ENT-E1 establish on two different projections of the conjugate Weil pair. The *admissible pair*

*entropy* lives on the rank-3 admissible projection  $\Pi_q$ : spectral atomicity [84] fixes  $\text{ran}(\Pi_q^{(c)}) = \text{span}\{e_0, e_{\xi_c}, e_{q-\xi_c}\}$  and the mean-zero lemma removes the constant mode, leaving the two-dimensional Born–Infeld parity pair  $\{e_{\xi_c}, e_{q-\xi_c}\}$  identified with the  $\text{spin-}\frac{1}{2}$  representation; the singlet identification of  $Q\beta$  [91] then gives  $S_{\text{ent}}^{\text{adm}} = \log 2$  for general canonical blocks. This is the entropy of the state behind the singlet correlator and the unconditional Tsirelson bound [22]. The *full residual-rank entropy*  $S_{\text{ent}}(n) = \log r_{\text{pair}}(n)$  lives on the full Gram–Schmidt basis (rank  $q$  at saturation, not rank 3) and is a pipeline-level contraction observable, unconditional for matched blocks (ENT-E1) and otherwise conditional on the residual admissibility indiscernibility hypothesis  $[H_{\text{res}}]$ ; the meta-plectic dilation of  $O\beta 0$  [76] is excluded as a bridge for  $[H_{\text{res}}]$  (**proved**;  $[H_{\text{res}}]$  for general canonical blocks: **open**).

**SchwingerLamb** [24]: Fine-structure constant  $\alpha$  as a projective invariant; Lamb shift and Schwinger correction rederived in the pre-geometric framework (**structural**; quantitative radiative corrections: **heuristic**).

## 5.5 Cosmological and Condensed-Matter Applications

**Cosmology** [2]: Flat galaxy rotation curves and the Hubble tension from a common effective mechanism; no dark matter or modified gravity (**structural**; quantitative fit: **heuristic**).

**Superconductivity** [3]: Projective phase locking as the structural origin of superconducting coherence across BCS, cuprate, and nickelate systems (**structural**).

## 6 Dependency Structure

Figure 1 presents the logical dependency DAG. Three entry paths emerge for external readers.

**Foundational path.**  $\text{ENI} \rightarrow \text{Foundation} \rightarrow \text{HeisenbergStructure} \rightarrow \text{noscale}$ . Requires least prior knowledge; establishes the axiomatic core.

**Spectral path.**  $\text{BornInfeld} \rightarrow \text{SpAdm} \rightarrow \text{SpRel} \rightarrow \text{O1} \rightarrow \dots \rightarrow \text{O24} \rightarrow \text{O25} \rightarrow \text{O28} \rightarrow \text{O29}$ . Develops the spectral machinery and the transfer chain.

**Physical path.**  $\text{Foundation} + \text{O18} + \text{O22} + \text{O23} \rightarrow \text{Q1} \rightarrow \text{Q2} \rightarrow \text{Q3}$ ;  $\text{Q2} \rightarrow \text{Q5a} \rightarrow \text{Q5b} \rightarrow \text{Q6a} + \text{Q6b}$ ;  $\text{O23} + \text{O28} + \text{O29} + \text{Q5b} \rightarrow \text{Q7}$  (dimensional bridge); parallel:  $\text{Spectral2} \rightarrow \text{Gravity} \rightarrow \text{Thermodynamics} \rightarrow \text{Lorentz}$ ;  $\text{Q5b} + \text{Q11} + \text{PTO} + \text{BI} \rightarrow \text{LorCap} \rightarrow \text{TempProj} \rightarrow \text{LCII} \rightarrow \text{LCIIo1}$ ;  $\text{LCo1}$ ,  $\text{LCo3}$ .

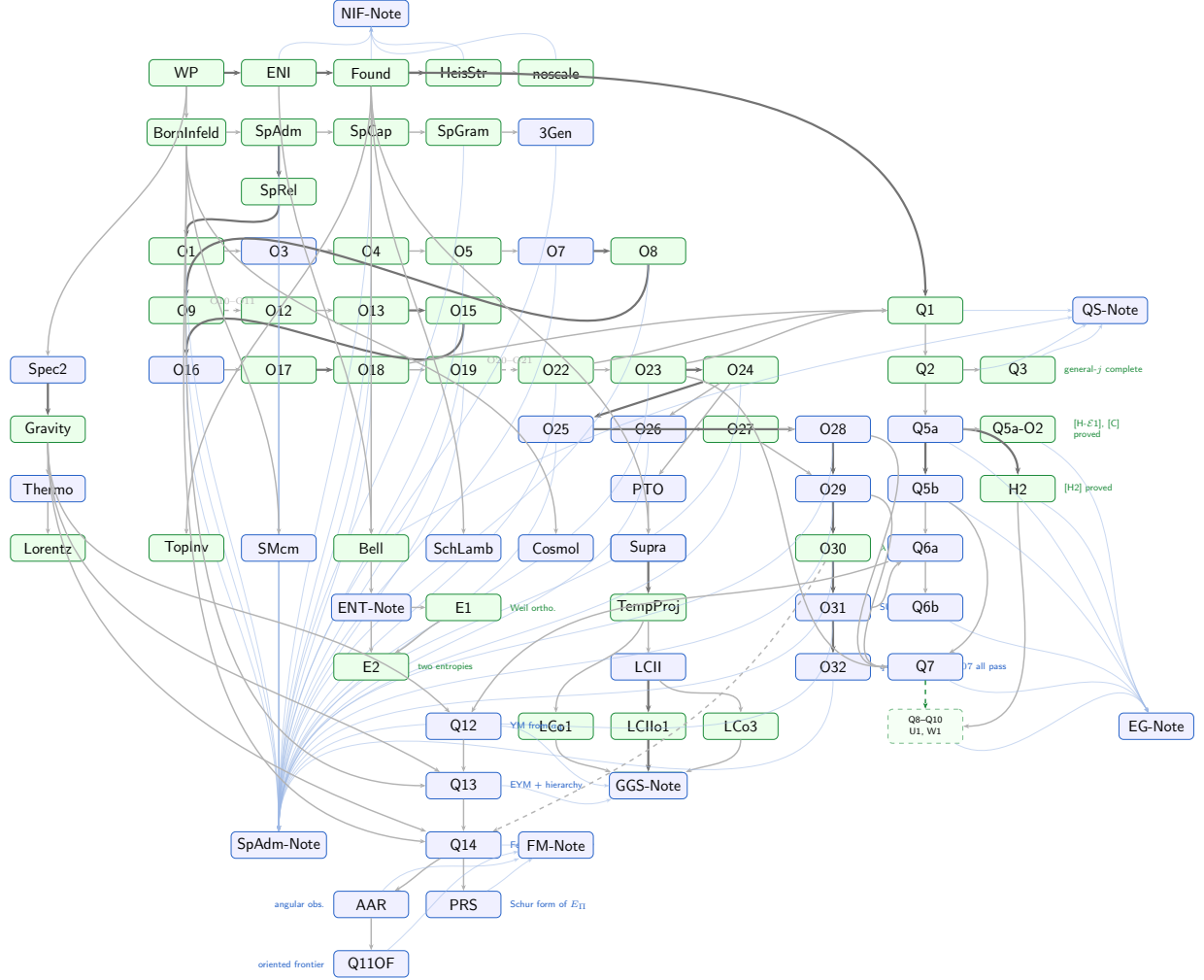


Figure 1: Logical dependency DAG of the Cosmochrony programme. Colour coding: ■ **proved**, ■ **structural**, ■ **heuristic**, ■ **open**. Thick arrows indicate the main logical spine; thin arrows indicate supporting dependencies. Dashed arrows with labels abbreviate linear sub-chains. The faint converging edges gather the constituent papers of each thematic sub-programme into its synthesis node: the Branch I foundational papers (ENI, Foundation, HeisStr, noscale) into **NIF-Note**, the spectral admissibility papers into **SpAdm-Note**, the Lorentz-capacity papers into **LC-Synth**, the geometric emergence papers into **EG-Note**, Q1/Q2/Q3 and the Bell paper into the quantum-structure **QS-Note**, Q12 and Q13 into the gauge-gravity stratification **GGs-Note**, and Q14 with the companion notes PRS, AAR and Q11OF into the fermionic-matter **FM-Note**. The dashed green arrow from Q7 leads to the geometric emergence closure block Q8–Q10 and U1, detailed in Figure 2. The interactive version with clickable nodes is provided as a companion HTML file.

Table 1 lists all papers in the programme. The **Key** column gives the cosmochrony-bibliography citation key.

| Label      | Short title                      | Key  | Main contribution                                                                                                                                                                                                 | Status     |
|------------|----------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| WP         | White Paper                      | [15] | Programme overview; substrate $\chi$ ; iterated projection; BI constraint; relaxation cascade                                                                                                                     | structural |
| ENI        | Emergence & non-injectivity      | [34] | No-go: genuine emergence $\Leftrightarrow \Pi$ non-injective; projection entropy $S_\Pi$                                                                                                                          | proved     |
| Found      | Foundation                       | [6]  | Axioms A1–A4; derives $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ and $V_\rho$ as theorems (Thm 5.6); v1.10: co-metric $g^{\mu\nu} = \text{diag}(-2, 2, 2, 2)$ noted; relation to HJ, symplectic, WKB, FRG formalisms | proved     |
| HeisStr    | Heisenberg structure             | [33] | Detailed proof of Found. Thm 5.6; uncertainty principle from A3 alone                                                                                                                                             | proved     |
| noscale    | No external scale                | [32] | No independent dimensional parameter beyond $c_\chi$                                                                                                                                                              | proved     |
| BornInfeld | Born–Infeld geometry             | [12] | BI action as unique minimal bounded-flux representation                                                                                                                                                           | proved     |
| SpAdm      | Spectral admissibility ( $Q_8$ ) | [52] | $A_n^{\max} = c_\chi/\sqrt{\lambda_n}$ ; first tractable instance                                                                                                                                                 | proved     |
| SpCap      | Spectral capacity                | [53] | Capacity functional $\mathcal{C}(G, S)$ ; binary-polyhedral maximality proved via adjoint identity $\hat{A}_{\text{adj}} = 2M - I$ and icosahedral isotropy                                                       | proved     |
| SpGram     | Gram rigidity (v1.0.1)           | [37] | Neutral-generator geometry from $G_{st} = -\frac{1}{2}\chi_\rho(st)$                                                                                                                                              | proved     |
| 3Gen       | Three generations                | [68] | Three-generation structure from ADE stratigraphic levels                                                                                                                                                          | structural |
| SpRel      | Spectral relaxation              | [8]  | $\Lambda_{\text{proj}} \asymp \lambda_2$ for LPS expander families                                                                                                                                                | proved     |
| O1         | Variable-valence graphs          | [43] | LPS saturation is a fixed-valence artefact; growing valence restores window                                                                                                                                       | proved     |
| O3         | Mass hierarchy                   | [41] | First SM connection: $\beta^* \in (0.09, 0.13)$ from lepton mass ratios                                                                                                                                           | structural |
| O4         | Upper bound $\beta \leq 1$       | [57] | BI flux + Cheeger $\Rightarrow \beta \leq 1$                                                                                                                                                                      | proved     |
| O5         | Rep-theoretic obstruction        | [5]  | Fixed-repr. approaches cannot close the $\beta$ gap                                                                                                                                                               | proved     |
| O6         | Matrix redundancy                | [30] | Matrix-level confinement of cascade exponent                                                                                                                                                                      | proved     |
| O7         | Continuum target                 | [39] | Target: $\delta \in [7.4, 10.6]$ on 3-step fingerprints                                                                                                                                                           | structural |
| O8         | LPS geometric obstruction        | [1]  | Exponential shell growth $\Rightarrow$ usable window $= O(\log q)$ BFS steps                                                                                                                                      | proved     |
| O9         | Heisenberg graphs                | [40] | Structural transition: $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ with $ B_n  \sim n^4$ replaces LPS                                                                                                                 | proved     |

*continued...*



| Label | Short title                              | Key  | Main contribution                                                                                                                                                                                                                                                  | Status     |
|-------|------------------------------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| O10   | Large- $q$ computa-<br>tion              | [38] | Confirms $n^4$ growth; TensorSketch ob-<br>struction identified                                                                                                                                                                                                    | proved     |
| O11   | Fourier-character<br>proxy               | [73] | Bidimensional proxy overcomes Ten-<br>sorSketch obstruction                                                                                                                                                                                                        | structural |
| O12   | Exact Weil blocks                        | [17] | $\hat{\delta}_{\text{exact}} \approx 4.3\text{--}4.8$ for $q \leq 211$ ; final ob-<br>struction resolved                                                                                                                                                           | proved     |
| O13   | Extended prime<br>range                  | [9]  | $\hat{\delta}(q) : 4.80 \rightarrow 3.59$ ; finite-size hypothe-<br>sis refuted                                                                                                                                                                                    | proved     |
| O14   | Observable-class<br>correction           | [36] | Corrects $\delta \mapsto \beta^*$ via block normalisa-<br>tion                                                                                                                                                                                                     | structural |
| O15   | Block-level no-go                        | [35] | No reweighting of individual blocks<br>reaches $\delta \in [7.4, 10.6]$ (Cor. 4.8)                                                                                                                                                                                 | proved     |
| O16   | Pair observable                          | [18] | $\sigma_{\text{pair}} = \sigma_c \sigma_{q-c}$ ; exponent doubling<br>$\delta_{\text{pair}} \approx 7.44$                                                                                                                                                          | structural |
| O17   | Structural deriva-<br>tion of pair       | [19] | $\rho_{q-c} = \bar{\rho}_c$ forced by fibre admissibility:<br>pair is derived, not postulated                                                                                                                                                                      | proved     |
| O18   | BI parity involution                     | [31] | Minimal fibre condition $\{\chi, -\chi\}$ ; BI<br>parity is the unique global symmetry<br>of $S_{\text{BI}}$                                                                                                                                                       | proved     |
| O19   | Canonical normali-<br>sation             | [4]  | $r(c, q)$ does not affect $\delta_{\text{pair}}$ ; observable<br>pipeline-independent                                                                                                                                                                              | proved     |
| O20   | Persistence criterion                    | [42] | Dynamical selection of admissible<br>modes via projective persistence                                                                                                                                                                                              | structural |
| O21   | Shell saturation                         | [51] | Observable rank $n_3^{\text{obs}}$ via $\Sigma_c(n_3) = 3$ ;<br>transfer $\delta_{\text{pair}} \rightarrow \beta^*$                                                                                                                                                | structural |
| O22   | Projection locking                       | [50] | BI projection locking forces $n_3^{\text{obs}}$ onto a<br>BFS shell (Thm 3.2)                                                                                                                                                                                      | proved     |
| O23   | Quaternionic mini-<br>mality             | [69] | $\Sigma_c(n_3) = 3$ from quaternionic maximal-<br>ity of neutral generators                                                                                                                                                                                        | proved     |
| O24   | Observable rank<br>rigidity              | [72] | Verticality Lemma; unconditional<br>transfer $c_\chi \rightarrow \delta_{\text{pair}} \rightarrow \beta^* \approx 0.126$<br>(Cor. 4.2)                                                                                                                             | proved     |
| O25   | Numerical campaign                       | [58] | $\bar{\delta}_{\text{pair}} \in \{8.89, 9.98, 9.55, 9.13, 8.78\}$ for<br>$q \in \{29, 61, 101, 151, 211\}$ ; inter-pair<br>$\sigma_q : 0.54 \rightarrow 0.11$ ; $\delta_{\text{corr}} \in [7.7, 9.0] \subset$<br>$[7.4, 10.6]$                                     | structural |
| O26   | Quadratic comple-<br>tion                | [45] | Dictionary: Weil blocks $\leftrightarrow$ rank-1 ma-<br>trices in $M_2(\mathbb{C})$ ; connects $\text{Heis}_3$ to $2I \subset$<br>$\text{SU}(2)$                                                                                                                   | structural |
| O27   | Quaternionic rigid-<br>ity               | [47] | Every admissible $\Phi_{q,\rho}$ satisfying three<br>structural constraints factors through<br>$\mathfrak{su}(2)$                                                                                                                                                  | proved     |
| O28   | Window calibration<br>& $r_{\text{eff}}$ | [7]  | $n_1(q)/q$ decreasing (Scenario S2);<br>$\delta_{\text{corr}} \in [7.4, 10.6]$ confirmed; $r_{\text{eff}} = 3$<br>in $\text{End}(H_{\text{eff}})$ , eigenstructure $[1 : \frac{1}{2} : \frac{1}{2}]$<br>invariant across all pairs and<br>$q \in \{61, 101, 151\}$ | structural |

*continued...*

| Label      | Short title                            | Key  | Main contribution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Status     |
|------------|----------------------------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| O29        | Symmetric rank formula & sector ID     | [55] | BI parity forces $M_j \in \text{Sym}(V_\rho, \mathbb{C})$ ; $r_{\text{eff}} = d_\rho(d_\rho + 1)/2$ ; unique inversion at $r_{\text{eff}} = 3$ gives $d_\rho = 2$ (spin- $\frac{1}{2}$ ); confirmed for $q \in \{61, 101, 151, 211\}$ ; $q = 101$ anomaly resolved by multi-sample analysis                                                                                                                                                                                                                                                      | structural |
| O30        | Eigenvalue ratio $\lambda_1/\lambda_2$ | [76] | BI parity forces equatorial constraint $ \alpha_j  =  \beta_j $ ; phase-independent central component gives $\mathcal{C}_c = \langle r^4 \rangle \text{diag}(1/4, 1/2, 1/4)$ ; ratio $[1 : \frac{1}{2} : \frac{1}{2}]$ derived analytically; factor of 2 is purely structural (Carnot degree-2 of $Z = [X, Y]$ )                                                                                                                                                                                                                                 | proved     |
| O31        | SU(3) from colour triplets             | [85] | Hurwitz obstruction ( $d_\rho = 3$ cannot arise from single-fibre argument, proved); colour triplets exist iff $q \equiv 1 \pmod{3}$ (proved); metaplectic intertwining proves $\sigma_c = \sigma_{\omega c}$ on $\mathcal{S}_q^{(3)}$ (proved, Thm 4.4); single-frequency fingerprint structure (Lem. 4.22) proves $[H\text{-color}]_{\text{pointwise}}$ exactly on the standard graph (Prop. 4.23); SU(3) is the unique triplet symmetry group (proved); $G_{\text{SM}}$ derived unconditionally for $\mathcal{S}_q^{(3)}$ and $\mathcal{S}_q$ | structural |
| O32        | [H-color] numerical test               | [77] | [H-color] passes for all $q \in \{61, 151, 211, 307\}$ ( $R \leq 3$ , $\text{ctrl\_ref} = 1.657 \times 10^{-3}$ ; $R = 2.1, 0.4, 1.0, 0.7$ respectively); $C_{\text{color}}$ rank 1 in all cases; $\delta_{\text{tri}} \approx 3\delta_c$ sub-percent; $[H\text{-color}]_{\text{avg}}$ proved to $O(q^{-1})$ ; $[H\text{-color}]_{\text{pointwise}}$ proved analytically in O31 (Prop. 4.23)                                                                                                                                                     | structural |
| PTO        | Temporal ordering                      | [44] | $I(n) = \sigma_{\text{pair}}(0) - \sigma_{\text{pair}}(n)$ is a non-decreasing temporal proxy; confirmed for $q \in \{29, 61, 101, 151, 211\}$                                                                                                                                                                                                                                                                                                                                                                                                   | structural |
| FibreErase | Fibre erasure                          | [80] | Monotone $I(c; \sigma(\ell))$ on BFS coarse-graining axis $\ell$ via DPI (Thm 4.1, conditional on [H-suff]); [H-suff] proved at the rank level (Prop. 3.6); capacity-level numerical evidence for $q \in \{61, 151, 211\}$ ; explicitly not a $c$ -theorem                                                                                                                                                                                                                                                                                       | structural |
| Q1         | Phase coherence                        | [22] | Phase coherence from admissibility (Thm 2.7); singlet correlator and Tsirelson bound derived                                                                                                                                                                                                                                                                                                                                                                                                                                                     | proved     |
| Q2         | Beyond spin- $\frac{1}{2}$             | [46] | Spin- $\frac{3}{2}$ quantum structure; SU(2) as admissibility fixed point                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | proved     |

*continued...*

| Label  | Short title                        | Key  | Main contribution                                                                                                                                                                                                                                                                                                                                                                                                     | Status     |
|--------|------------------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Q3     | Universal spin- $j$ singlet        | [91] | Schur + BI indiscernibility identify proto-state as singlet $ \Omega_j\rangle$ for all $j \in \{\frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}\}$ ; universal correlator and Born rule derived; closes Q2 open step                                                                                                                                                                                                      | proved     |
| Q5a    | Large- $q$ operator limit          | [26] | Mosco convergence $\mathcal{E}_q \rightarrow L_\Pi = -A\partial_x^2$ ; conditional on [H2]. v2.0: [H1] established on admissible sector (not needed in full generality); [H- $\mathcal{E}1$ ] and [C] proved by Q5a-O2 [84]; [H2] proved by H2 paper: all hypotheses verified on the admissible sector                                                                                                                | structural |
| Q5a-O2 | Spectral atomicity & Mosco closure | [84] | Admissible sector spectrally atomic: $\text{ran}(\Pi_q^{(c)}) = \text{span}\{e_0, e_{\xi_c}, e_{q-\xi_c}\}$ , $\xi_c/q \rightarrow \omega_c \neq 0$ . Proves [H- $\mathcal{E}1'$ ] (scaled coercivity, $\beta = 1$ optimal) and [C] (Mosco compactness via Bolzano–Weierstrass in $\mathbb{C}^2$ , no Nash inequality needed). Numerical confirmation $K_q = 1$ at machine precision for $q \in \{29, 61, 101, 151\}$ | proved     |
| Q5b    | 4D Lorentzian geometry             | [11] | BFS stratification of $\text{Heis}_3(\mathbb{R}) \rightarrow 4\text{D}$ Lorentzian spacetime; $g^{\mu\nu} = 2\eta^{\mu\nu}$ fully determined (Q9, Q10, Q11 close all open problems)                                                                                                                                                                                                                                   | proved     |
| Q6a    | Gauge structure                    | [23] | U(1) and SU(2) from fibre fixed points; SU(3) $\times$ SU(2) $\times$ U(1) derived (O31 [85], proved for $\mathcal{S}_q$ via [H-color] <sub>pointwise</sub> )                                                                                                                                                                                                                                                         | structural |
| Q6b    | Effective spacetime                | [16] | $g^{\mu\nu} = \text{diag}(-2, 2, 2, 2)$ from $\Pi_q$ ; Schwarzschild from flux conservation; Einstein equations as consistency conditions; closes chain $\Pi_q \rightarrow \mathcal{L}_\Pi \rightarrow g^{\mu\nu} \rightarrow G_{\mu\nu}$ ; [H-lift] closed by Q9                                                                                                                                                     | structural |
| Q7     | Dimensional bridge                 | [89] | $H_{\text{eff}} \cong \text{Sym}^2(V_\rho)$ ; $[F_c, L_{\text{Weil}}] = 0$ ; bridge non-obstruction (Case 3 excluded by Q9); $A_z = C_{\text{su}(2)} = 2$ ; uniqueness of $\varphi$ up to scalar                                                                                                                                                                                                                      | structural |
| Q8     | Casimir rigidity                   | [56] | $A_Z = C_{\text{su}(2)} = 2$ unconditionally under Q5a; $\lambda$ fixed by horizontal sector consistency; $g^{\mu\nu} = \text{diag}(-A_\tau, 2, 2, 2)$ established                                                                                                                                                                                                                                                    | structural |
| Q9     | [H-lift] proved                    | [81] | Kinetic-sector identification: modulation generator suppressed at rate $O(q^{-1/2})$ ; [H-lift] is a theorem under Q5a; $A_H > 0$ proved by coercivity, independently of bridge                                                                                                                                                                                                                                       | proved     |

*continued...*

| Label               | Short title                      | Key      | Main contribution                                                                                                                                                                                                                                                                                                                                                    | Status     |
|---------------------|----------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Q10                 | Asymptotic isotropy              | [75]     | $A_H(q) \rightarrow 2$ with $ A_H - 2  \leq C \varepsilon(q)$ ; universality [U] $\Rightarrow \mathfrak{su}(2)$ -isotropy $\Rightarrow$ Casimir; bridge conjecture resolved (isotropic case) under [U]                                                                                                                                                               | structural |
| U1                  | Universality proved              | [U] [90] | $\varepsilon(q) = O(q^{-1/2})$ from Weil-generator Lipschitz bound and BFS–Carnot–Carathéodory convergence; no BI input required for small- $\theta$ regime                                                                                                                                                                                                          | proved     |
| W1                  | [H-w] proved                     | [92]     | $a_q(s) \rightarrow A > 0$ uniformly in $s$ : fibre-invariance of admissible observables ( $\mathcal{O} = \text{Im } \Pi$ ) is the structural origin; U1 controls the rate $O(q^{-1/2})$ ; after W1 + Q5a-O2 [84], open Q5a hypotheses reduced to [H2] only                                                                                                          | proved     |
| H2                  | [H2] proved                      | [83]     | Strong convergence $\hat{X}_q \rightarrow x$ , $\hat{P}_q \rightarrow -i\partial_x$ on $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ : quantitative Poisson aliasing lemma (rate $O(q^{-1})$ ), discrete Sobolev identity, quasi-isometry of sinc embedding; closes last open hypothesis of Q5a Theorem 7.1                                                                | proved     |
| Q11                 | Temporal Casimir rigidity        | [86]     | Closes Q5b open problem O3: Schur rigidity forces $A_\tau = C_{\mathfrak{su}(2)} = 2$ ; effective Lorentzian co-metric fully determined: $g^{\mu\nu} = \text{diag}(-2, 2, 2, 2) \propto \eta^{\mu\nu}$                                                                                                                                                               | structural |
| Q12                 | Yang–Mills from $a_4$            | [93]     | Seeley–DeWitt $a_4 \supset \frac{1}{12} \text{Tr}(F^2)$ ; vertical variation $\delta_{\mathcal{A}} S_\Pi = 0$ yields $D_\mu F^{a\mu\nu} = 0$ ; induced coupling $g_{\text{YM}}^{-2} \sim \log(\Lambda/\mu)$ ; gauge–gravity hierarchy structural (proved for $a_4$ derivation; conditional on Q6a for $G_\Pi$ )                                                      | structural |
| Q13                 | Gauge–gravity spectral synthesis | [79]     | Joint $\delta_{g,\mathcal{A}} S_\Pi = 0$ yields coupled Einstein–Yang–Mills system; $a_6$ cross-coupling $R_{\mu\nu\rho\sigma} F^{\mu\nu} F^{\rho\sigma}$ (suppressed by $\ell_{\text{sp}}^2$ ); Eddington–Born–Infeld joint completion; hierarchy ratio $G_N g_{\text{YM}}^2 \sim \ell_{\text{sp}}^2 / (\dim V) \cdot [\log(\Lambda/\mu)]^{-1}$ without fine-tuning | structural |
| <i>continued...</i> |                                  |          |                                                                                                                                                                                                                                                                                                                                                                      |            |

| Label               | Short title                                 | Key  | Main contribution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Status     |
|---------------------|---------------------------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Q14                 | Fermionic matter & chirality                | [78] | Admissible spinor bundle $S_\Pi$ reproduces $SU(2)_L \times U(1)_Y$ ; projected Dirac endomorphism $E_\Pi$ enforces $V-A$ chirality via the spinorial BI parity lift (proved); $\gamma_5$ -weighted $a_4$ fixes hypercharge by anomaly cancellation; $\sigma_c(n_3) = 3$ yields three-generation factor $\mathbb{C}_{\text{gen}}^3$ ; quark colour sector unconditional at pointwise level ( $[H\text{-color}]_{\text{pointwise}}$ proved)                                                                                                                                         | structural |
| PRS                 | Schur form of $E_\Pi$ + localisation of $u$ | [88] | $E_\Pi = -\Pi_S D(1-P)D\Pi_S^* = -M^\dagger M$ Schur complement of eliminated sector; generation split $u$ controlled by $\mathcal{D}^\pm$ -transported antiunitary chiral equivariance defect $\Delta_\chi(P) = \pi_{LL} - \tau \overline{\pi_{RR}} \tau^{-1}$ ; A1–A3 force $u = 0$ , non-zero $u$ requires A4 chiral symmetry-breaking; finite/Lorentzian separation: $u_{\text{fin}} = 0$ for $q$ odd prime; operator $u = \text{spectral } u$ on flat effective metric; reduces open problem to Lorentzian chiral curvature $\mu_\chi^2 := \partial_s^2 \mathcal{B}(0)$       | structural |
| AAR                 | Angular amplitude observable                | [82] | BCH reduction identifies per-step $J_\Pi$ -odd $J_3$ component of metaplectic cascade generator with oriented symplectic area swept per step (Carnot-degree-2 increment of $\text{Heis}_3$ ); observable is accumulated oriented $J_3$ -odd area along Weil–BFS cascade normalised by $\widehat{I}(n)$ , equals $\varepsilon_{\text{Weil}}(n_3^{\text{obs}})$ to be compared with dictionary $\varepsilon = 1/10$ without fitting; O12 fingerprint: $B_c$ carries radial $\sigma_c$ , $A_c$ carries angular split; symmetric BFS shells give $\Theta_{\text{raw}} = 0$ identically | structural |
| Q11OF               | Oriented frontier observable                | [87] | Symmetric-capacity obstruction makes raw shell observable vanish identically ( $g \mapsto g^{-1}$ pairs opposite central phases); frontier transfer observable on directed outgoing edges $g \rightarrow gs$ with central increment Heisenberg cocycle $a(g)s_b$ (discrete BCH $\frac{1}{2}[X, Y]$ ); orientation is recursive arrow $\partial_\tau = \log g$ ; first test is existence of oriented signal $\langle \Delta A_c \rangle_{\partial+S_m} \neq 0$ ; anti-bias control via symmetrised-frontier cancellation                                                            | structural |
| <i>continued...</i> |                                             |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |            |



| Label    | Short title                                                   |          | Key  | Main contribution                                                                                                                                                                                                                                                     | Status     |
|----------|---------------------------------------------------------------|----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Spec2    | Relational time                                               | space-   | [48] | Operational metric from graph Laplacians; spacetime from spectral correlations                                                                                                                                                                                        | structural |
| Gravity  | IR Einstein response + BI UV completion                       |          | [25] | $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}^{(\Pi)}$ from $\delta S_{\Pi}/\delta g$ at leading IR order; unique BI determinantal UV completion                                                                                                                                  | proved     |
| Thermo   | Spectral thermodynamics                                       |          | [28] | Local spectral first law; Einstein equation as spectral equilibrium condition                                                                                                                                                                                         | structural |
| Lorentz  | Gravitational waves                                           |          | [13] | Lorentzian completion; causal propagation; GW spectrum                                                                                                                                                                                                                | proved     |
| LorCap   | Lorentz factor from capacity                                  |          | [54] | $\eta_n^\tau \rightarrow \beta\gamma$ ; $\Phi(\eta_n^\tau) \rightarrow 1/\gamma$ ; BI factorization and central-phase factorization ( <b>proved</b> ); $\mathcal{P}_\tau^{\text{Q5b}}$ construction: closed by TempProj [59]                                          | structural |
| TempProj | Temporal map                                                  | residual | [59] | $\mathcal{P}_\tau^{\text{Q5b}} \Sigma_n^{\text{tot}} = \sqrt{1 - B_n^2}$ ; uniqueness (three-input necessity); Lemma 1 of LorCap proved unconditionally (cond. on [U])                                                                                                | proved     |
| LCII     | Fibrewise lapse                                               | capacity | [20] | Fibrewise extension to non-homogeneous regime; $d\tau/dn _x = R(x)/\gamma$ ; unified capacity arbitration: $1/\gamma$ (kinematic) and $R(x)$ (gravitational) as two facets of the same BI trade-off                                                                   | structural |
| LCIIo1   | Local spectral closure of the capacity-metric bridge          |          | [27] | $R(x)^2 = 1 - \epsilon(x) = -2g^{\tau\tau}(x)$ derived from local admissibility data alone; SU(2)-Schur reduction $A_\tau(x) = 2(1 - \epsilon(x))$ combined with the principal-symbol identification $g^{\tau\tau}(x) = -A_\tau(x)/4$ ; no Einstein equation required | proved     |
| LCo1     | Lorentz transformations from projective temporal ordering     |          | [29] | Effective Lorentz transformations between inertial observers are uniquely fixed by the projective temporal capacity load $\beta$ ; length contraction $\ell = \ell_0/\gamma$ as a corollary                                                                           | proved     |
| LCo3     | Finite- $q$ corrections to the Carnot capacity identification |          | [21] | Parabolic Carnot formula deviates from the exact torus capacity by $\Delta_q(n) = 4m^2/q^2$ ( $n = R + m$ , $R = (q - 1)/2$ ); quadratic identity exact, Carnot identification holds for $n \leq R$                                                                   | proved     |
| TopInv   | Topological invariants                                        |          | [70] | Charge quantisation ( $\pi_1 \cong \mathbb{Z}$ ), no monopoles, chirality (Hopf fibration)                                                                                                                                                                            | proved     |
| SMcm     | Charge, mass, inertia                                         |          | [14] | Mass and charge as BI-saturated responses                                                                                                                                                                                                                             | structural |
| Bell     | Bell non-locality                                             |          | [10] | Bell violations from non-injectivity alone                                                                                                                                                                                                                            | proved     |

*continued...*

| Label   | Short title              | Key  | Main contribution                                                         | Status     |
|---------|--------------------------|------|---------------------------------------------------------------------------|------------|
| SchLamb | Fine-structure constant  | [24] | $\alpha$ as projective invariant; Lamb shift and Schwinger correction     | structural |
| Cosmol  | Rotation curves / Hubble | [2]  | Flat rotation curves and Hubble tension from a common effective mechanism | structural |
| Supra   | Superconductivity        | [3]  | Projective phase locking as origin of superconducting coherence           | structural |

## 7 Open Problems

### 7.1 Branch II (O-Series)

**Extension of O29 to larger primes.** The symmetric rank formula is confirmed for  $q \in \{61, 101, 151, 211\}$  (105/105 pairs at  $q = 211$ , 100%, multi-sample confirmed). Extension of the numerical computation to  $q \in \{307, 401\}$  is pending.

**Non-conjugate block protocol.** A measurement protocol using non-conjugate block data (e.g. the  $c$ -block at two independent primes) would probe the full rank  $d_\rho^2 = 4$  and provide a direct test of O26 Criterion 5.4 in its original formulation.

**Analytical control of  $n_1(q)/q$ .** The BFS window ratio  $n_1(q)/q$  decreases monotonically through  $q = 211$  (O28, Scenario S2); the asymptotic constant  $\hat{\alpha}$  is bounded below by 0.062 (updated from O28 with  $q = 211$  data) but not yet stabilised.

**Notation consolidation:**  $c_\chi \rightarrow c_{\text{BI}}$ . A coordinated rename of the BI saturation constant across all papers is planned and deferred pending consistency with already-deposited O-series papers.

### 7.2 Branch III (Q-Series and Applications)

**Bridge existence at finite  $q$  (Q7–Q10).** The asymptotic isotropy  $A_H \rightarrow 2 = A_Z$  is proved by Q10 and U1 (conditionally on [U]), establishing the isotropic limit of Q7 Conjecture 4.7. What remains open is the finite- $q$  structural question: whether an explicit  $\mathfrak{su}(2)$ -equivariant isomorphism  $\varphi_q : \text{Sym}^2(V_\rho) \xrightarrow{\sim} W_{\text{sp}}$  exists at each prime  $q$ , not only in the limit.

**$\pi_2(\mathcal{C}_{\text{eff}}) = 0$  beyond the canonical model (TopologicalInvariants, Remark 4.2).** General proof of this topological property is open.

**Full SM gauge group: [H-color] and O32 (Q6a, O31).** O31 [85] derives  $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$  as a composite of admissibility fixed points unconditionally for the colour-adapted Cayley graph and, with [H-color] proved analytically (Proposition 4.23 of O31), unconditionally for the standard graph as well. O32 [77] confirms [H-color] numerically for all four test primes  $q \in \{61, 151, 211, 307\}$ : all pass  $R \leq 3$  (noise floor `ctrl_ref` =  $1.657 \times 10^{-3}$ , controls  $q \in \{29, 101\}$ ;  $R = 2.1, 0.4, 1.0, 0.7$  respectively). What remains open: analytical derivation of [H-color] from Foundation axioms; transfer chain  $\delta_{\text{tri}} \rightarrow \alpha_s$  (strong coupling constant); identification of quarks as stable projected configurations carrying colour quantum numbers.

**Baryon-to-photon ratio (SMchargemass).** Quantitative derivation of  $\eta \sim 10^{-10}$  is open.

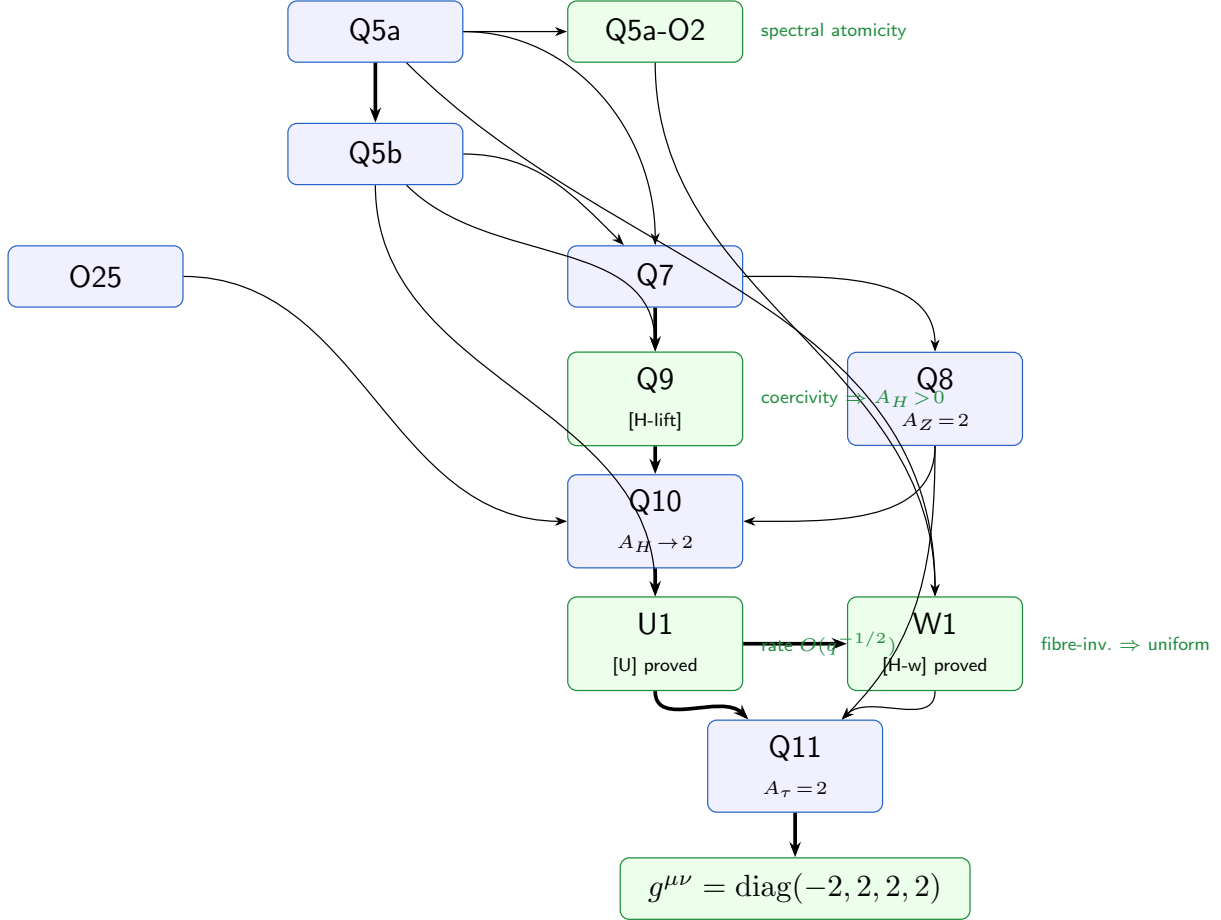


Figure 2: Geometric emergence sub-programme: logical chain from Q5a to the effective Lorentzian co-metric. **Q5a-O2** [84] (**proved**): spectral atomicity of  $\text{ran}(\Pi_q)$ ; proves [H- $\mathcal{E}$ 1] and [C]; closes all Q5a hypotheses except [H2]. **Q9** (**proved**): [H-lift] is a theorem under Q5a; coercivity gives  $A_H > 0$  independently of the bridge. **Q8** (**structural**):  $A_Z = C_{\mathfrak{su}(2)} = 2$  by Casimir rigidity; normalisation fixed by horizontal sector. **Q10** (**structural**): universality [U]  $\Rightarrow \mathfrak{su}(2)$ -isotropy  $\Rightarrow A_H \rightarrow 2$ , conditional on [U]. **U1** (**proved**): [U] proved with rate  $O(q^{-1/2})$  from Weil-generator Lipschitz bound and BFS–Carnot–Carathéodory convergence (Q5b). **W1** (**proved**): [H-w] proved as a structural consequence of fibre-invariance ( $\mathcal{O} = \text{Im } \Pi$ ), quantitative control from U1. The effective metric  $g^{\mu\nu} = \text{diag}(-A_\tau, 2, 2, 2)$  is established under the Q5a hypotheses; [H2] is the sole remaining open hypothesis.

**$V - A$  chiral structure and fermionic matter (Q14).** Q14 [78] addresses the formal derivation of the  $V - A$  coupling: the projected Dirac endomorphism  $E_\Pi$ , via the spinorial lift of BI parity (now proved), is left-admissible and enforces chiral selection. Hypercharge is not a free parameter but is fixed by anomaly-cancellation constraints on the  $\gamma_5$ -weighted  $a_4$  Seeley–DeWitt coefficient. The three-generation multiplicity follows from  $\sigma_c(n_3) = 3$ . A qualitative mechanism for the inter-generation splitting is now identified (Q14 §6: static obstruction, oriented  $J_\Pi$ -odd lift, non-zero cascade  $J_3$  projection). Residual open points: the amplitude of the splitting (the cascade normalisation  $\varepsilon = \mathcal{N}_{\text{casc}}|\alpha|$ , to be derived from  $c_\chi \rightarrow \delta_{\text{pair}} \rightarrow \beta^*$ ), the explicit mass generation mechanism, generation mixing via the complex metaplectic phase, and the Lorentzian extension of the projected Dirac operator.

**Gauge–gravity synthesis (Q12  $\rightarrow$  Q13).** Q12 establishes Einstein gravity and Yang–Mills dynamics as independent ( $a_2$  and  $a_4$ ) sectors of the single functional  $S_\Pi[g, \mathcal{A}]$ . Q13 [79] completes this synthesis: the joint variational problem  $\delta_{g, \mathcal{A}} S_\Pi = 0$  yields the coupled Einstein–Yang–Mills system, the precise hierarchy ratio  $G_N g_{\text{YM}}^2 \sim \ell_{\text{sp}}^2 / (\dim V) \cdot [\log(\Lambda/\mu)]^{-1}$ , and the Eddington–Born–Infeld completion of the joint functional. Residual open points: EBI tensor  $\Xi_{\mu\nu}$  ambiguity (subleading), numerical  $a_6$  coefficients, Lorentzian extension of the joint EBI functional, and extension to include projected matter currents  $J^{a\nu}$ .

**Yang–Mills uniqueness for SU(3) (Q12, O31).** Q12 derives the Yang–Mills equations for any compact  $G_\Pi$  from the  $a_4$  heat-kernel coefficient; the SU(3) sector is inherited from O31/Q6a. What remains open is the internal SU(3) uniqueness problem (O31 §9.2): showing that the SU(3) gauge dynamics emerge *uniquely* from the colour triplet co-admissibility structure, without inputting the group as data.

**Lorentz Capacity Sub-Programme.** The Lorentz-capacity sub-programme is now closed. TempProj closes the  $\mathcal{P}_\tau^{\text{Q5b}}$  construction of LorCap; LCHo1 [27] closes the spectral derivation of  $R(x)$  (formerly LC-O2-O1, see Remark 3.2 of [20]):  $A_\tau(x) = 2(1 - \epsilon(x))$  via SU(2)-Schur reduction on  $\text{Sym}^2(V_\rho)$  combined with  $g^{\tau\tau}(x) = -A_\tau(x)/4$  yields  $R(x)^2 = 1 - \epsilon(x) = -2g^{\tau\tau}(x)$  without importing the Einstein equation. LCo1 [29] closes LC-O1: the effective Lorentz transformations between inertial observers are uniquely fixed by the projective temporal capacity load  $\beta$ , yielding length contraction  $\ell = \ell_0/\gamma$  as a corollary. LCo3 [21] closes LC-O3: the parabolic Carnot identification holds for  $n \leq R := (q - 1)/2$  with deviation  $\Delta_q(n) = 4m^2/q^2$  for  $n = R + m$ , while the quadratic identity  $B_n^2 + (F_n^\tau)^2 = 1$  is exact.

**Thermodynamics: Lorentzian extension and matter coupling.** The local spectral first law is derived in the Riemannian setting. Analytic continuation and the explicit form of  $W_\Pi$  for matter fields are open.

## 8 Conclusion

The Cosmochrony programme has established a coherent, multi-level derivation of physical structure from a single primitive. Branch I (Foundation, ENI, HeisenbergStructure [33]) is formally closed: the Weil representation of  $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$  is a theorem. The binary-polyhedral maximality conjecture (SpectralCapacity [53]) is proved:  $2I$  uniquely maximises  $\mathcal{C}_\alpha(G, S)$  for all  $\alpha > 0$  among finite  $\text{SU}(2)$  subgroups with neutral generating sets, closing the precursor programme for the O-series. Branch II (O-series) has closed the unconditional transfer chain  $c_\chi \rightarrow \delta_{\text{pair}} \rightarrow \beta^*$  analytically (O24) and numerically validated for  $q \in \{29, 61, 101, 151, 211\}$  (O25). The spin- $\frac{1}{2}$  sector identification is complete (O29): the symmetric rank formula  $r_{\text{eff}} = d_\rho(d_\rho + 1)/2$  uniquely identifies  $d_\rho = 2$  from the observed  $r_{\text{eff}} = 3$ , confirmed numerically for  $q \in \{61, 101, 151\}$ . The invariant eigenvalue ratio  $\lambda_1/\lambda_2 = 2$  of the per-pair covariance is now analytically derived (O30): the Born–Infeld parity anti-linearity forces an equatorial constraint on the admissible fibre, and the factor of 2 is purely structural—the squared symmetric-tensor normalisation of the Carnot-degree-2 central generator  $Z = [X, Y]$ . O31 develops the structural framework for  $\text{SU}(3)$ : the metaplectic intertwining of  $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$  forces triplet co-admissibility unconditionally for the colour-adapted Cayley graph, and  $\text{SU}(3)$  is thereby identified as the symmetry group of the co-admissible colour triplet, proved for the standard graph by Proposition 4.23 of O31 [85]. Branch III (Q-series and companions) derives quantum mechanics, spacetime, and SM observables at the structural level, with several results conditional on explicitly stated hypotheses. The  $\text{SU}(2)$  quantum sector is now complete: Q3 closes the general- $j$  open step of Q2 by identifying the proto-state as the singlet for all five admissible sectors of  $2I$  via Schur’s lemma, and derives the universal correlator  $E(\hat{a}, \hat{b}) = -j(j+1)/3 \cdot (\hat{a} \cdot \hat{b})$  and Born rule for  $j \in \{\frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}\}$  (**proved**). The geometric emergence sub-programme has advanced substantially: Q9 proves [H-lift] and  $A_H > 0$  (unconditionally under Q5a); Q10 proves  $A_H \rightarrow 2$  under spectral universality [U]; U1 proves [U] with rate  $O(q^{-1/2})$  from BFS convergence and Weil-generator stability; W1 proves [H-w] as a structural consequence of the fibre-invariance of admissible observables. The effective Lorentzian co-metric  $g^{\mu\nu} = \text{diag}(-2, 2, 2, 2) \propto \eta^{\mu\nu}$  ( $A_\tau = 2$  from Q11 [86]) is thereby established under the Q5a foundational hypotheses and [U]. Q12 [93] completes the spectral derivation of gauge dynamics: the Seeley–DeWitt  $a_4$  coefficient of the operator  $A_{g,\mathcal{A}}$  on the admissible vector bundle contains the Yang–Mills density  $\frac{1}{12}\text{Tr}(F^2)$ , and the vertical variation  $\delta_{\mathcal{A}}S_\Pi = 0$  yields  $D_\mu F^{a\mu\nu} = 0$ . Q13 [79] completes the gauge–gravity synthesis: the joint variational problem  $\delta_{g,\mathcal{A}}S_\Pi = 0$  yields the coupled Einstein–Yang–Mills system, and the gauge–gravity hierarchy  $G_N g_{\text{YM}}^2 \sim \ell_{\text{sp}}^2/(\dim V) \cdot [\log(\Lambda/\mu)]^{-1} \ll 1$  emerges as a structural consequence of the spectral stratification ( $a_2$  quadratic vs.  $a_4$  logarithmic UV divergence), without fine-tuning. Q14 [78] extends this to the fermionic sector: the admissible Weil module induces a spinor bundle  $S_\Pi$  reproducing  $\text{SU}(2)_L \times U(1)_Y$ ; the projected Dirac endomorphism  $E_\Pi$  enforces  $V-A$  chirality via the spinorial lift of BI parity (now proved); hypercharge is fixed by anomaly-cancellation constraints on the  $\gamma_5$ -weighted  $a_4$  coefficient; and the saturation invariant  $\sigma_c(n_3) = 3$  yields a gauge-singlet three-generation factor  $\mathbb{C}_{\text{gen}}^3$ . A qualitative mechanism for the inter-generation splitting is identified in Q14 §6 (a static  $J_\Pi$ -protected degeneracy lifted by the oriented cascade generator), with its amplitude deferred to the cascade normalisation. Together, Q12–Q14 establish that the bosonic gauge and gravitational fields and the fermionic matter content of the Standard Model arise from a single structural source: the non-injective projection admissibility constraint on the Weil module. The Cosmochrony programme can now be stated in one sentence: it is a *spectrally stratified theory of emergent interactions*, in which the type of a fundamental interaction is determined by the Seeley–DeWitt order at which the admissible operator responds. Q5a v2.0 establishes that [H1] is not needed in full generality on the admissible sector. Q5a-O2 [84] proves spectral atomicity of the admissible sector and closes [H- $\mathcal{E}1$ ] and [C]: each conjugate pair contributes exactly three pure Fourier modes, coercivity at the optimal diffusive scale follows from the non-zero limiting frequency, and Mosco compactness reduces to Bolzano–Weierstrass in  $\mathbb{C}^2$ . The H2 paper proves the last open hypothesis [H2] (strong convergence of the rescaled Weil generators  $\hat{X}_q \rightarrow x$  and

$\hat{P}_q \rightarrow -i\partial_x$ ) via a quantitative Poisson aliasing lemma and a discrete Sobolev identity; all hypotheses of Q5a Theorem 7.1 are verified on the admissible sector, and the large- $q$  convergence theorem holds unconditionally.

For an external reader, three entry sequences are suggested. For the quantum structure thread: ENI  $\rightarrow$  Foundation  $\rightarrow$  O18  $\rightarrow$  O22  $\rightarrow$  O23  $\rightarrow$  Q1. For the spectral mass hierarchy thread: ENI  $\rightarrow$  Foundation  $\rightarrow$  O16  $\rightarrow$  O24. For the spacetime thread: Spectral2  $\rightarrow$  Gravity  $\rightarrow$  (Thermodynamics or Lorentz); for the geometric emergence thread: Q5a  $\rightarrow$  Q5b  $\rightarrow$  Q7  $\rightarrow$  Q8  $\rightarrow$  Q9  $\rightarrow$  Q10  $\rightarrow$  U1  $\rightarrow$  W1  $\rightarrow$  H2  $\rightarrow$  Q11.



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Einstein–Yang–Mills system  $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}^{\text{YM}}$  from the joint variation (Q13 Thm 3.2, structural);  $a_6$  gauge–gravity cross-coupling  $R_{\mu\nu\rho\sigma} F^{\mu\nu} F^{\rho\sigma}$  (Q13 Thm 4.1, structural); Eddington–Born–Infeld joint completion (Q13 Thm 5.3, conditional on [H-ext]); structural hierarchy ratio  $G_N g_{\text{YM}}^2 \sim \ell_{\text{sp}}^2 / \dim V \cdot [\log(\Lambda/\mu)]^{-1} \ll 1$  without fine-tuning (Q13 Prop. 6.1). Q12 is shared with the gauge-structure sub-programme (Note 3). Open:  $a_6$  phenomenological normalisation; derivation of [H-ext] from A1–A4; coupled equations with fermionic matter currents from Note 6, doi:[10.5281/zenodo.20563349](https://doi.org/10.5281/zenodo.20563349).

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